A Costly-information Foundation for Psychometric Curves*

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Abstract

This paper studies how the cost of distinguishing similar situations shapes the pattern of mistakes in binary choices. I consider an agent who decides how much effort to invest in learning about a continuous state before choosing between two actions whose payoffs depend on the state. In perceptual tasks, the state can represent the relative strength of two stimuli. Building on the Fisher information cost introduced by Hébert and Woodford (2021), I characterize the optimal state-dependent choice rule and give conditions under which it generates an S-shaped psychometric curve. Furthermore, I derive comparative statics showing how incentives and prior beliefs alter the response curve. In symmetric tasks, stronger incentives reduce mistakes, while greater concentration of the prior near the decision boundary increases errors. More generally, an action is chosen more often at every state when its relative payoff rises or when beliefs shift toward states that favor it. The results provide a costly-information foundation for psychometric curves and link their shape to the economic environment.

JEL Codes: D83, D91, C61

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1 Introduction

Many important decisions rely on evidence that is costly to acquire or interpret. A lender decides whether to approve a loan after assessing a borrower's repayment prospect; a physician chooses a treatment after evaluating diagnostic evidence; and a judge rules on a case after reviewing a potentially complex record. In each case, the decision maker can also choose whether to acquire more information before acting, by requesting extra financial records, ordering additional tests, or scrutinizing the record more carefully. Such efforts may improve the decision, but require time and cognitive effort. The resulting choice may depend on the stakes of the decision and the costs of processing evidence, and understanding how they jointly shape choices is important.

Rational inattention provides a framework for studying this trade-off. Rather than assuming the decision maker's information is exogenously given, it allows her to choose how much information to acquire before acting. This approach has been applied to problems ranging from price setting and portfolio choice to strategic interaction (see Maćkowiak, Matějka, and Wiederholt, 2023, for a review). Its flexibility, however, makes the specification of the information costs central to its behavioral implications. Much of the literature measures cost by Shannon mutual information, the expected reduction in uncertainty about the state. Linear Shannon costs yield a tractable connection to logit choice in discrete-action problems (Matějka and McKay, 2015).

Despite its analytical appeal, Shannon mutual information can make overly restrictive predictions. Consider a perception experiment, in which a subject sees a display of 100 dots in either blue or red, and receives a fixed reward for each correct choice of which color forms the majority. A display with 51 red dots is much harder to classify than one with 90, although choosing red yields equal rewards in both cases. Under the linear Shannon costs, the optimal response prescribes choosing red with the same probability under in these two cases, and actually in every display with a red majority. (See Figure 1 for illustration.) This reflects a general feature of Shannon cost: states with identical action payoffs are treated equally (Caplin, Dean, and Leahy, 2022).[1]

These restrictions are at odds with experiments findings. In perceptual discrimination tasks, subjects typically make more errors for stimulus that are closer to the categorical boundary. Moreover, the probabilities of a given response often traces an S-shaped psychometric curve (Wichmann et al., 2018). Such curve exhibits diminishing sensitivity away from the decision boundary: a change in relative stimulus strength has a smaller effect on choice probabilities when the evidence already strongly favors one response. In an incentivized classification experiment, Dean and Neligh (2023) find that subjects make fewer errors for stimuli farther from the category boundary, even though the reward for correct classification is un-

[1] This follows from the *Invariance under Compression* property of Shannon model studied by Caplin et al. (2022). It also implies that shifting prior probability between displays with 51 and 90 red dots leaves the optimal response unchanged whenever the total prior probability of a red majority is held fixed.

changed.[2] These findings motivate an information-cost model that accounts for perceptual distance.

In this paper, I provide a costly-information explanation for why choice behavior exhibits diminishing sensitivity away from the decision boundary. I study a binary choice problem over an ordered state space, in which the decision maker decides how finely to distinguish the nearby states before acting. The state may represent the relative strength of a sensory stimulus, or repayment prospects, likelihood of a disease, re-offending risk, and so on. I model the the information cost using *Fisher information cost* introduced by Hébert and Woodford (2021). Under this specification, I characterize the optimal response rule and show how it changes with incentives and prior beliefs

The cost function is based on Fisher information, which measures how sensitively a probability distribution changes with respect to the underlying parameter. Here, the parameter is the state, and the relevant distribution is the decision maker's stochastic choice rule at that state. The Fisher information cost aggregates the local sensitivities across state space, making it costly for choice probabilities to vary sharply across nearby states.

My first contribution is a characterization of the shape of optimal response rule. In a symmetric environment, where relabeling the the two actions and mirroring the state around the threshold leaves the problem unchanged, I establish a necessary and sufficient condition for the optimal response to be sigmoidal at every incentive level (Proposition 5). The condition is that the *Residual incentive ratio* is nonincreasing in the distance to the threshold. The Residual incentive ratio measures the incentive pressure to continue adjusting behavior as the state becomes more extreme. In a fixed-prize discrimination task, this condition is satisfied by nondecreasing hazard rate of the stimulus distribution, including uniform and distributions with log-concave density. A complementary result establishes sigmoidality of the response curve under a uniform prior and a nondecreasing payoff difference, without requiring symmetry (Proposition 6).

These results explain the familiar S-shape of psychometric curves as an optimal response to costly discrimination, rather than imposing a particular functional form. They help to clarify the limit of predictions: symmetry alone does not guarantee sigmoidality, and sufficiently nonmonotone payoffs can generate more complex responses even under a uniform prior. The model also yields a distinct prediction about where choices are most informative about the underlying state. Under a uniform prior and a strictly single-crossing payoff, the Fisher information of an informative optimal response peaks at the cutoff where the preferred action switches. This cutoff need not coincide with the stimulus level at which the two actions are chosen with equal probability. In the standard Gaussian signal-detection model with linear stimulus encoding and constant noise variance, by contrast, the Fisher information of the binary choice peaks at the 50-percent choice threshold, regardless of the decision criterion.

My second contribution is a set of comparative statics describing how the choice rule re-

[2] In a separate incentive experiment, they find that accuracy improves substantially less with higher rewards than predicted by linear Shannon model.

sponds to changes in decision environments. When the two actions are treated symmetrically, raising the incentives steepens the response curve at the cutoff and reduces error at every state (Proposition 8). The model also yields quantitative predictions on how fast accuracy improves: Suppose the decision maker receives a fixed reward U for each correct classification and the underlying difficulty of distinguishing nearby states does not change. The average error rate shrinks towards zero at a rate of $1/\sqrt{U}$ as the stakes becomes higher (Proposition 9). This is in contrast with the exponential decline predicted by linear Shannon costs. Beyond symmetric changes in incentives, the model predicts that raising the relative payoff of an action at some states increases the probability of it being chosen at every state, including states whose payoffs are unchanged (Proposition 10).

I derive further comparative statics that separate the effects of incentives from shifts in the environment. In the symmetric setting, holding payoffs and the underlying difficulty of discrimination fixed, concentrating a symmetric prior towards the cutoff in the likelihood-ratio sense flattens the response curve: slope at the cutoff falls and error probabilities at every state increase (Proposition 11). Unlike a decline in the average accuracy, this change cannot directly be explained by difficult cases getting sampled more often. Meanwhile, when the prior is shifted toward the states where an action is favored in the likelihood ration sense, the choice probability of that action raises at every state (Proposition 12).

These results yield testable restrictions on how choices respond to changes in the decision environment. The sigmoidality of the response curve alone does not distinguish the Fisher-information model from other explanations of noisy choice. A fitted logit or probit function can describe behavior in a given environment, but does not by itself determine how its parameters should change with incentives or the distribution of stimuli. By contrast, holding the underlying discrimination difficulty fixed, the Fisher-information model predicts the entire response curve under new payoffs and priors using a cost parameter calibrated in a baseline condition. These restrictions can be taken to data with a standard discrimination task. A laboratory design would repeatedly present continuous stimuli with a known correct classification, vary independently the reward for accuracy and the distribution of stimuli, and estimate the subject's response curve in each condition. With the cost parameter estimated from the baseline and held fixed, such a design can test three predictions: whether higher rewards reduce errors at every stimulus level, and whether average errors decline at the predicted rate; whether concentrating stimuli near the cutoff raises errors at physically identical stimuli; and whether, under asymmetric rewards, discrimination is finest at the payoff cutoff rather than at the 50-percent choice point.

A key analytical difficulty is that the Fisher information cost couples response at every state to its neighbors. The decision problem cannot be solved separately at each state, and the cost becomes singular as choice probabilities approach zero or one. I address these difficulties with a geometric reparameterization that converts the Fisher information term into a quadratic term on the derivative of an angular variable. I establish existence of an optimal response and uniqueness whenever it is informative, characterize when the decision maker

acquires information, and derive an Euler–Lagrange boundary-value equation with natural endpoint conditions. Under a uniform prior, the equation has a simple mechanical interpretation as the motion of a pendulum under a state-dependent force. This characterization provides the basis for the smoothness, curvature, and comparative-statics results. Hébert and Woodford (2021) established the neighborhood-based foundation and related qualitative response properties; the analysis here sharpens the shape restrictions and develops their consequences for observable behavior under changes in payoffs and priors. It also connects with other state-sensitive information costs (Pomatto et al., 2023; Bloedel et al., 2026).

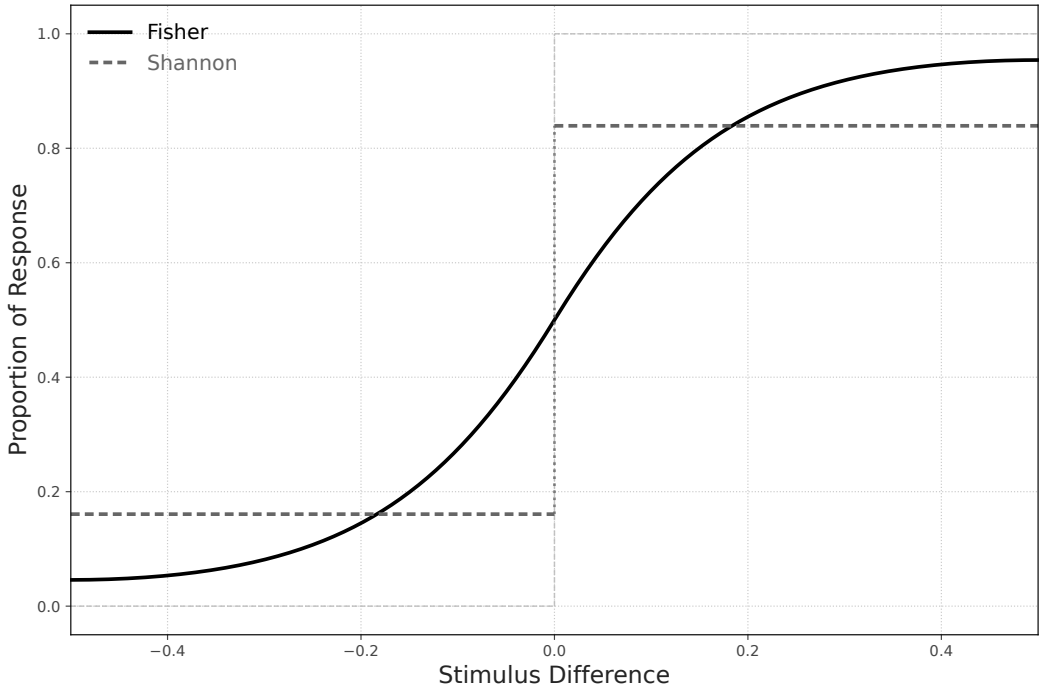


Figure 1: Optimal Response Curves and Simulated Experimental Fits

1.1 Outline of the paper

The remainder of the paper is organized as follows. Section 2 introduces the binary-choice problem and the Fisher information cost. Section 3 develops the behavioral implications, including the decision to acquire information, conditions for sigmoidal response curves, and comparative statics with respect to incentives and prior beliefs. Section 4 provides the mathematical foundations: a geometric reparameterization and the associated first-order equation that characterizes the optimal response, it establish existence and uniqueness of the optimal solution, and yields a criterion for whether the decision maker acquires information.

2 The Model

A decision maker is choosing between two alternatives L and R, whose payoffs depend on the underlying state $\theta \in \Theta$. The state space is assumed to be a compact interval $\Theta = [a, b]$ in $\mathbb{R}$. The decision maker is uncertain about the state and is endowed with full support prior with density g. Her task is to choose an optimal signal structure that maximizes her expected utility minus the cost of generating the signal structure.[3]

Formally, a signal structure consists of a pair $\langle S, \sigma \rangle$. We use S to denote the signal space, which is assumed to be finite.[4] Meanwhile, $\sigma : \Theta \to \Delta(S)$ is a measurable mapping from the state space to distributions over the signal space, with $\sigma(\theta) := \sigma(\cdot|\theta)$ being interpreted as the signal distribution conditional on the state being θ. For the information cost to be well-defined later, we also assume that, for each $s \in S$, the map $\theta \mapsto \sigma(s|\theta)$ is absolutely continuous on Θ.[5] The collection of signal structures that satisfies the above assumption is denoted by $\mathcal{S}$. After observing a signal $s \in S$, the decision maker chooses an action to maximize expected utility under the posterior induced by s.

The decision maker can flexibly select any signal structure from the collection $\mathcal{S}$ at a cost $C : \mathcal{S} \to [0, +\infty]$. Specifically, we assume that the cost function is the *Fisher information* cost (Hébert and Woodford, 2021, Section II.C), which is the averaged Fisher information. The Fisher information quantifies informativeness by measuring how much the signal distributions varies across states locally. It is formally defined as the variance of the partial derivative of the log-likelihood with respect to θ, called the score:[6]

$$\mathcal{I}_F(\{\sigma(\theta)\}) := E_{s|\theta}\left[\left(\frac{\partial}{\partial\theta}\log\sigma(s|\theta)\right)^2\right] = \sum_{s:\sigma(s|\theta)>0}\sigma(s|\theta)\left(\frac{\partial}{\partial\theta}\log\sigma(s|\theta)\right)^2.$$

The Fisher information cost is:

$$C_{FI}(\sigma; g) := E_\theta[\mathcal{I}_F(\{\sigma(\theta)\})] = \int_\Theta g(\theta)\mathcal{I}_F(\{\sigma(\theta)\}\,d\theta = \int_\Theta g(\theta)\sum_{s:\sigma(s|\theta)>0}\sigma(s|\theta)\left(\frac{\partial}{\partial\theta}\log\sigma(s|\theta)\right)^2 d\theta.$$

The Fisher information cost satisfies Blackwell monotonicity: a more informative signal structure, in the sense of Blackwell (1951), incurs a higher cost. We prove this in Ap-

[3] By a *signal structure* we mean what is often called a *Blackwell experiment*. We use "signal structure" instead of "experiment" to avoid confusion with laboratory experiments.

[4] So $\Delta(S)$ is a $(|S|-1)$-simplex and the topology on it is the Euclidean topology from $\mathbb{R}^{|S|}$; The topology on Θ is the Euclidean topology on $\mathbb{R}$.

[5] A function y is absolutely continuous on an interval $[a, b]$ if and only if there exists a Lebesgue-integrable function f on $[a, b]$ such that it can be represented as

$$y(t) = y(a) + \int_0^t f(\tau)d\tau$$

Hence, an absolutely continuous function y has a derivative almost everywhere.

[6] Although a common support assumption is standard, we allow $\sigma(s|\theta) = 0$. Since S is finite and $\sigma(s|\theta)$ is absolutely continuous for each s, the score has mean zero almost everywhere, so Fisher information can still be interpreted as the variance of score.

pendix C.1. Consequently, we can restrict attention to signal structures that provide direct action recommendations. To see this, consider any signal structure $\langle S, \sigma \rangle \in \mathcal{S}$ and a decision rule $q : S \to \Delta(A)$. We can merge all signals that induce the same action and define

$$\hat{\sigma}(a|\theta) = \sum_{s \in S} q(a|s)\sigma(s|\theta), \text{ for each } a \in A.$$

The merged signal structure leads to the same state-contingent choice rule, and thus the same expected payoff as the original one. Furthermore, since $\hat{\sigma}$ is a Blackwell garbling of σ, Blackwell monotonicity of C_{FI} implies $C_{FI}(\hat{\sigma}; g) \leq C_{FI}(\sigma; g)$. Lastly $\hat{\sigma}$ remains in $\mathcal{S}$ because S is finite and each $\sigma(a|\cdot)$ is a finite sum absolutely continuous functions. Therefore, with binary choice, we may take the signal space to be $\{L, R\}$ and interpret each signal as a direct recommendation of the corresponding action.

We use the short-hand notation $p(\theta) := \sigma(R|\theta)$ to denote the probability of receiving a signal recommending action R in state θ. We also normalize the state-dependent utility of choosing action L to be 0 and denote the utility of selecting R in state θ as $u(\theta)$.[7] We require p to be absolutely continuous, and denote the set of such state-dependent stochastic choice rules by $AC(\Theta, [0,1]) := \{p : \Theta \to [0,1] \,|\, p \text{ is absolutely continuous}\}$.

In the binary signal setting, the Fisher information cost can be rewritten as:

$$C_{FI}(p; g) = \int_\Theta g(\theta) \frac{(p'(\theta))^2}{p(\theta)(1 - p(\theta))} \, d\theta$$

where we treat

$$\frac{(p'(\theta))^2}{p(\theta)(1-p(\theta))} = \begin{cases} \frac{(p'(\theta))^2}{p(\theta)(1-p(\theta))}, & \text{if } p(\theta) \in (0,1) \\ 0, & \text{if } p'(\theta) = 0 \text{ and } p(\theta) = 0 \text{ or } 1 \\ +\infty & \text{other cases} \end{cases}.$$

Specifically, we allow the decision maker to acquire no information at no cost, that is, when p is a constant, thus totally uninformative, the cost is 0.

The gross expected payoff under the choice rule p is $\int_\Theta g(\theta)u(\theta)p(\theta)\, d\theta$, and the net expected payoff is:

$$\int_\Theta g(\theta)u(\theta)p(\theta)\, d\theta - \lambda C_{FI}(p; g),$$

where $\lambda \in (0, +\infty)$ is a scaling parameter that can be interpreted as scaling the information cost, or equivalently, the incentive level.

In summary, the decision maker needs to choose a state-dependent stochastic choice rule

[7] This normalization is without loss of generality when the decision maker has von Neumann expected utility, since the expected payoff differs by a constant that is independent of the choice rule:

$$\int g(\theta)[\sigma(R|\theta)u_R(\theta) + (1 - \sigma(R|\theta))u_L(\theta)] = \int g(\theta)[\sigma(R|\theta)(u_R(\theta) - u_L(\theta))] + u_L(\theta).$$

to solve the following optimization problem:

$$\underset{p\in AC(\Theta,[0,1])}{\text{maximize}} \int_{\Theta} g(\theta)\left[u(\theta)p(\theta) - \lambda\frac{(p'(\theta))^2}{p(\theta)(1-p(\theta))}\right]d\theta \tag{P}$$

Maintained assumptions. Unless stated otherwise, we maintain the following regularity and nondegeneracy conditions throughout. These assumptions are standard and primarily technical, and additional restrictions on the payoff and prior are introduced when they are first used in Section 3.

Assumption 1 (Payoff regularity)**.** The payoff function u is piecewise continuous, that is, for $\Theta = [a, b]$, there exists a finite partition $a = \theta_0 < \theta_1 < \cdots < \theta_N = b$, such that

1. u is continuous on (θ_i, θ_{i+1}) for $i = 0, 1, \cdots, N-1$;
2. left limits $\lim_{\theta\to\theta_i^-} u(\theta)$ exist and are finite for $i = 1, 2, \cdots N$, and right limits $\lim_{\theta\to\theta_i^+} u(\theta)$ exist and are finite for $i = 0, 1, \cdots N-1$.

Assumption 2 (Prior regularity)**.** The prior admits a density g on $\Theta = [a, b]$ with respect to Lebesgue measure. The density satisfies $g(\theta) > 0$ for all θ, g is absolutely continuous, and $\int_a^b g(\theta)\,d\theta = 1$.

Assumption 3 (Nondegeneracy)**.** The payoff function u differs from zero on a set of positive measure.

Since the state space Θ is a compact interval, Assumption 1 implies u is bounded and integrable, while Assumption 2 implies g is bounded away from zero. Assumption 3 rules out the degenerate case where the decision maker is indifferent in almost every state,. Finally, because the prior g is strictly positive, a set of states has positive prior probability if and only if it has positive Lebesgue measure. Hence, references to sets of positive measure may refer to either measure.

3 Behavioral implications

This section derives the implications of Fisher information cost on observable choices. We will answer the following questions in turn: when the decision maker choose to pay attention (Section 3.1), what shape the response curve takes (Section 3.2), and how the response curve shifts in response to changes in incentives and prior beliefs (Section 3.3). The proofs can be found in Appendix A, and the mathematical results they build on are described in Section 4.

Throughout, we use the interval $[a, b]$ to refer to the state space Θ and call the midpoint $c = (a+b)/2$. We say that a prior g is symmetric if $g(c-\theta) = g(c+\theta)$, and a payoff function u is anti-symmetric if $u(c-\theta) = -u(c+\theta)$. A payoff is single-crossing if there exists some θ_u such that $u \le 0$ on $[a, \theta_\mu)$ and $u \ge 0$ on $(\theta_\mu, b]$; and it is strictly single-crossing if both inequalities are strict.

3.1 When does the decision maker pay attentions?

The decision maker can choose to completely ignore the evidence and result in a choice rule that is independent of and hence constant across states. We say that the decision maker *acquires information* if her choice rule has positive cost, $C_{FI}(p; g) > 0$, or equivalently, if p is not constant. Our first two results give simple conditions under which information acquisition happens: the decision maker always pays attention if she is ex-ante indifferent between the two options; if one option does not dominate the other in almost all states, then the decision maker will learn as long as the stake is high enough, or equivalently, when learning is cheap enough.

Proposition 1 (Ex-ante indifference implies attention). *Suppose Assumptions 1–3 hold. If neither action is preferred ex-ante:*

$$\int_a^b g(\theta)u(\theta)\,d\theta = 0,$$

then the decision maker acquires information for every $\lambda > 0$.

Proposition 2 (Strong incentive induces attention). *Suppose Assumptions 1–3 hold, and fix a prior g. If neither action dominates the other in almost every state:*

$$\textit{both } \{\theta : u(\theta) > 0\} \textit{ and } \{\theta : u(\theta) < 0\} \textit{ have positive measure,}$$

then there exists $\bar{\lambda} > 0$, such that the decision maker acquires information whenever $\lambda < \bar{\lambda}$. Equivalently, the decision maker acquires information under the payoff $A \cdot u$ for all sufficiently large $A > 0$.

Both results follow as corollaries of Theorem 4 in Section 4.4. The intuition again comes from a marginal benefit-cost argument. ... When the two actions have the same expected payoff, the decision maker can stay uniformed and choose each action with probability one-half at every state. She can also slightly tilt her response curve toward the better action in each state. The resulting payoff gain is first order in the size of this adjustment, while the incurred cost is second order. Hence, for any finite cost parameter λ, a sufficiently small adjustment raises net expected payoff. On the other hand, when neither action is dominated, a choice rule that is responsive to the state gains positive payoff, and increasing the stakes scales up its payoff advantage while leaving its information cost unchanged. The gain therefore exceeds the cost once incentives are sufficiently strong.

The conditions also apply to the linear mutual information cost. What is special to the Fisher information cost is that the underlying structure of the states affects the attention decision. Under mutual information cost, this decision depends only on the payoffs and the arrangement of states irrelevant. However, under Fisher information, the arrangement can be critical. Because staying responsive across a narrower or more fragmented set of states requires sharper choice probability changes, which are more costly. This is illustrated in the

following example.

Example 3.1. Let the prior be uniform on $[0,1]$. Suppose action L gives a higher payoff by one unit in 60% of states and while action R is better by one unit in the remaining 40% of the states. The payoff difference between the two actions is one.

In each arrangement, the state space $[0,1]$ is divided into n equal-sized blocks and action R is better on the top 40 percent of each block. Table 1 reports the largest cost parameter $\bar{\lambda}$ at which the decision maker acquires information.

Under the Fisher information cost, fragmenting the payoffs into n blocks lowers the threshold $\bar{\lambda}_F$ by a factor of approximately n^2. Equivalently, the stakes required to induce attention must rise by a factor of n^2. Conversely, under linear mutual information, the threshold $\bar{\lambda}_S$ is invariant to the arrangement.[8]

△

Number of blocks n	1	2	5	10
Fisher information cost: $\bar{\lambda}_F$	0.098	0.024	0.0039	0.00098
Linear mutual information cost: $\bar{\lambda}_S$	2.47	2.47	2.47	2.47

Table 1: Largest cost parameter at which the decision maker acquires information in Example 3.1.

Consider a portfolio manager selecting bonds. A manager who profits from buying any bond above a single, absolute credit score cutoff will study the financial disclosures. In contrast, a manager seeking to pick the top 40% of bonds within every distinct credit rating tier (e.g., the best AAs, the best As, the best Bs) faces a much harder task and might eventually give it up. This is a behavioral counterpart of the observation of Pomatto et al. (2023) that, when nearby states are hard to distinguish, learning whether a quantity is above a fixed cutoff is much cheaper than learning whether it is even or odd.

3.2 What is the shape of response curve?

We say that a response curve is *S-shaped* if it increases smoothly, in the sense of having no kinks, becoming steeper up to a single inflection point and flatter afterwards. Behaviorally, it means that, the decision maker's sensitivity to the state, as measured by changes in choice probability per unit change of state, first rises and then falls, peaking at a single interior point. More formally, it is defined as follows:

Definition 3.1. A function $p : [a,b] \to [0,1]$ is *sigmoidal* (or *S-shaped*) if $p \in C^1([a,b])$, p is strictly increasing, and there exists a $\tau \in (a,b)$ such that p' is strictly increasing on $[a,\tau]$ and strictly decreasing on $[\tau,b]$.

[8]The absolute values of $\bar{\lambda}_F$ and $\bar{\lambda}_S$ are not directly comparable; what matters is how each parameter responds to changes in the arrangement.

As a first step towards our S-shape results, we first show that the Fisher information costs implies monotonicity and smoothness under very general conditions.

Proposition 3 (Smoothness). *Under Assumptions 1–3, every optimal response p^* is continuously differentiable on $[a, b]$ and twice differentiable almost everywhere.*

A discontinuity in the payoff shows up only in the curvature of the response, but not in its value or slope. Intuition comes from the optimality condition: it gives a second-order semilinear boundary-value problem, with a natural pendulum analog (Section 4.2.1). Just as an abrupt change in external force alters acceleration at once while position and velocity evolve smoothly, a jump in payoffs shifts the second derivative without creating kinks or jumps in the choice probabilities. Quantitatively, the sensitivity of choice to the state is bounded by the ratio of incentive relatively to cost level:

$$\left|(p^*)'(\theta)\right| \leq \frac{\|u\|_\infty}{16\lambda g(\theta)} \qquad \text{for every } \theta.$$

Under a linear mutual information cost, the optimal choice rule is a logit in the payoff difference (Matějka and McKay, 2015):

$$p^S(\theta) = \frac{q\, e^{u(\theta)/\lambda_S}}{q\, e^{u(\theta)/\lambda_S} + 1 - q}, \tag{1}$$

where $q = \int g(\theta)\, p^S(\theta) d\theta$ is the unconditional probability of choosing R. Two features of this choice rule (1) stand out: the state enters only through the relative payoff $u(\theta)$, and the prior enters only through the scalar q. Consequently, the choice probability jumps when the payoff jumps. In the fixed-prize task, it is a step function (Figure 1). Beyond single-agent decision-making, continuity of the choice rule in the state also plays a crucial role in strategic environments: for instance, in global games, Morris and Yang (2022) show that continuity of choice in the state is a key ingredient ensuring equilibrium uniqueness.

Proposition 4 (Monotonicity). *Under Assumptions 1–3, suppose the payoff u is single–crossing, then every optimal response p^* is weakly increasing. Especially, if p^* is non-constant, then it is strictly increasing.*

The monotonicity result only requires the payoff to be single-crossing. As long as the state space can be partitioned into two regions, where L is better in the lower region and R is better in the upper one, stronger evidence in favor of R always increases the probability of choosing it. By contrast, under mutual information cost in (1), the choice rule p^S is is monotone in the state only if u itself is monotone.

3.2.1 Symmetric discrimination tasks

We call it a *symmetric environment* if the prior is symmetric and the relative payoff is anti-symmetric. In other words, the two actions are treated symmetrically, so relabeling them and mirroring the state around the midpoint leaves the problem unchanged.

In such a setting, the shape of the response curve is governed by the following ratio.

Definition 3.2 (Residual incentive ratio)**.** For $\theta \in [c, b]$, let

$$\mathcal{R}(\theta) := \frac{\int_\theta^b g(s)u(s)\,ds}{g(\theta)} = \frac{1 - G(\theta)}{g(\theta)} E[u(S) \mid S \geq \theta],$$

where G is the cumulative distribution function of the prior.

At a specific state θ, the term is defined as the expected payoffs in states more extreme than θ, relative to the likelihood of that state θ. It measures the incentive to continue adjusting behavior as the state moves farther into the tail.

The optimal response is sigmoidal at every incentive level if and only if the following term is nonincreasing as the state moves away from the midpoint.

Proposition 5 (S-shape in symmetric tasks). *Under Assumption 1–2, suppose the prior g is symmetric, the payoff u is antisymmetric and single-crossing at c,. Then for every $\lambda > 0$, the optimal response p^* is interior, with $p^*(c) = 1/2$. Moreover, p^* is sigmoidal for every scaling $\lambda > 0$ if and only if the residual incentive ratio $\mathcal{R}(\theta)$ is nonincreasing on $[c, b]$.*

In a symmetric fixed-prize discrimination task, the residual ratio $\mathcal{R}(\theta) = U \int_\theta^b g(s)\,ds/g(\theta)$ is the the prize U divided by the hazard rate of the prior. So a nondecreasing hazard rate of the stimulus distribution guarantees an S-shaped optimal response at every incentive level. In particular, this covers all the distributions with log-concave density, including Uniform, truncated Normal, truncated Logistic, and many other distributions.

Corollary 1 (Fixed-prize discrimination task). *Suppose Assumptions 1–3 hold, and the prior g is symmetric with nondecreasing hazard rate $g(\theta)/(1 - G(\theta))$, then under the antisymmetric fixed-prize payoff:*

$$u(\theta) = \begin{cases} -U, & \theta \in [a, c), \\ U, & \theta \in (c, b] \end{cases},$$

the optimal response is sigmoidal.

The condition also holds for every prior density g that is nondecreasing on $[c, b]$. Because the numerator of $\mathcal{R}$ is an integral with positive integrand and hence decreases on $(c, b]$, the ratio $\mathcal{R}$ is nonincreasing when the denominator increase.

Corollary 2 (U-shaped prior). *Under the assumptions of Proposition 5, suppose in addition that the prior density g is U-shaped, i.e., nonincreasing on $[a, c]$ and nondecreasing on $[c, b]$. Then, the optimal response p^* is sigmoidal.*

The following example demonstrates that our condition on the Residual incentive ratio cannot be dropped. Even in a symmetric environment with log-concave prior density, large incentives near extreme states can generate a response that is not S-shaped. It highlights the

boundary of our theory prediction and yields a testable empirical implication: introducing heavy tail bonuses will distort the classic sigmoidal response pattern. Such payoffs arise when extreme cases carry disproportionate stakes. For example, an auditor may be evaluated primarily on uncovering massive frauds or gross misstatements.

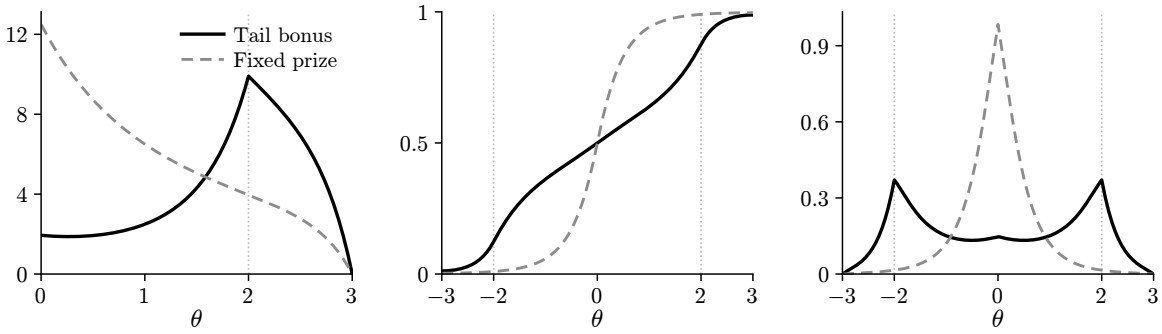
Example 3.2 (Large tail bonus brakes S-shape)**.** Suppose the prior distribution is the standard Normal distribution truncated to $[-3, 3]$, and suppose a correct choice earns 0.5 when $|\theta| < 2$ and 25 when $2 \le |\theta| \le 3$. The relative payoff is

$$u^{\text{bonus}}(\theta) = \operatorname{sgn}(\theta)\big(0.5 + 24.5 \cdot \mathbf{1}_{\{2 \le |\theta| \le 3\}}\big).$$

This payoff is antisymmetric, single-crossing, and nondecreasing, and the prior is symmetric with log-concave density.

The tail bonus raises the expected payoff beyond θ in the numerator of $\mathcal{R}$, while the prior density in its denominator falls as the state moves outward. Before the bonus threshold, the payoff remains small, allowing $\mathcal{R}$ to rise (Figure 2(a)).

By Proposition 5, p^*_λ is not sigmoidal for every sufficiently large λ. Figure 2 shows that the response is already non-sigmoidal at $\lambda = 1$. Its slope has a small peak at zero and two larger peaks at $\theta = \pm 2$. For comparison, a fixed-prize payoff $u^{\text{fixed}}(\theta) = 10\operatorname{sgn}(\theta)$ under the same prior produces an S-shaped response. △



(a) Residual incentive ratio $R(\theta)$ (b) Optimal response $p^*(\theta)$ (c) Slope $(p^*)'(\theta)$

Figure 2: Tail bonus and the shape of the optimal response.

3.2.2 Beyond symmetry tasks

In many real-life decisions, however, the two actions are treated asymmetrically. Missing a fraud may cost much more than a false alarm, and convicting an innocent may be worse than acquitting the guilty. Without symmetry, the S-shape of response curve survives under a uniform prior and monotone payoffs.

Proposition 6 (S-shape under monotone payoffs). *Under Assumptions 1–3, suppose the prior g is uniform, and the payoff u is nondecreasing. If the optimal response p^* is interior, then it is sigmoidal.*

With asymmetric payoffs, the state at which the two actions are chosen with equal probability need not coincide with the payoff cutoff. The next result says where the decision maker is most responsive to the state.

Proposition 7 (Where discrimination is finest). *Under Assumptions 1-2, suppose u is strictly single-crossing at θ_u and the optimal response p^* is interior.*

(i) The prior-weighted discriminability $g(\theta)\sqrt{I(\theta)}$ is strictly increasing on $[a, \theta_u]$ and strictly decreasing on $[\theta_u, b]$. In particular, under a uniform prior, the local Fisher information $I_F(\theta)$ is maximized at the payoff cutoff θ_u.

(ii) Suppose the prior is uniform and p^* crosses $1/2$ at $\theta_{1/2}$. Then $p^{*\prime}$ is strictly increasing on $[a, \min\{\theta_u, \theta_{1/2}\}]$ and strictly decreasing on $[\max\{\theta_u, \theta_{1/2}\}, b]$. Consequently, the peak of slope lies between θ_u and $\theta_{1/2}$.

Measured in Fisher information, the decision maker discriminates most finely exactly when the preferred action switches, wherever the curve happens to cross 50 percent. This distinguishes the model from a standard probit response, $\text{Probit}(\theta) = \Phi((\theta - \alpha)/\beta)$. Under a probit choice rule with a linear index and constant noise variance, Fisher information is maximized at the state α where the two actions are chosen with equal probability.[9] Under asymmetric payoffs the two models therefore disagree about where discrimination is finest: at the payoff cutoff, or at the point of subjective equality. This difference can be tested using repeated choices across stimulus levels, provided the payoff cutoff is known and the response curve can be estimated with sufficient precision.

Figure 3 illustrates the distinction. Suppose the prior is uniform on $[0, 1]$, R is the correct response in $(1/2, 1]$ where choosing it gives a reward of 180, compared with 60 for a correct L response in $[0, 1/2)$. The 50-percent choice point shifts to the left of the payoff cutoff. The peak of Fisher information, however, remains at the payoff cutoff. Panel (a) plots the optimal response, with the 50-percent point (dashed), the steepest point (dotted), and the payoff cutoff (solid). The gray dashed curve is a standard probit with the same 50-percent point and the same slope there. Panel (b) illustrates the root Fisher information $\sqrt{I_F(\theta)}$. Under the Fisher information cost, discrimination is finest at the payoff cutoff; for the probit response, it is finest at the 50-percent point.

[9] The same holds for a standard logit response.

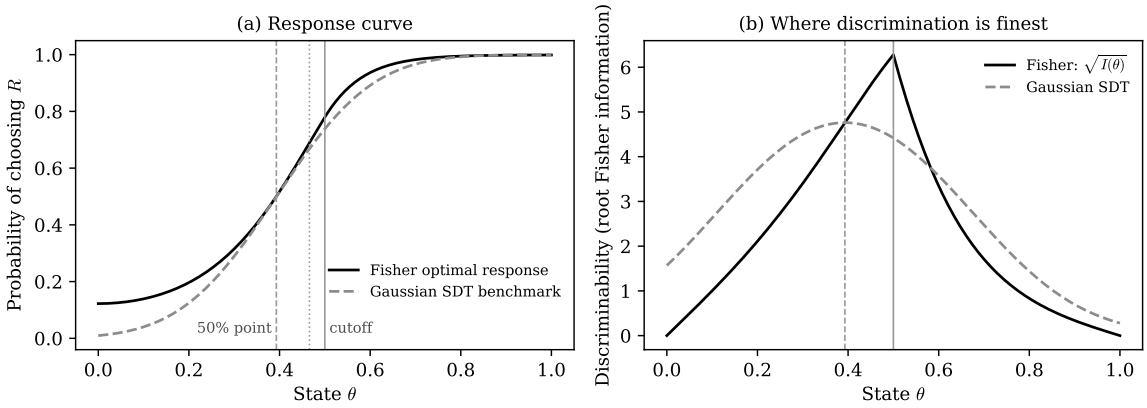


Figure 3: Asymmetric rewards and local discriminability.

3.3 How does the curve respond to environment changes?

Choice frequency data from perception experiments are usually summarized by a parametric specification (psychometric function) (Wichmann et al., 2018):

$$p(\theta) = \gamma + (1 - \gamma - \kappa)F(\theta; \alpha, \beta),$$

where F is some sigmoidal cumulative distribution function, e.g. Gaussian or Logistic distribution. The *shift parameter* α determines the horizontal position of the curve, and is usually represented by the threshold stimulus level at which the choice probability p reaches 50%; The *scale parameter* β determines the steepness of the curve, indicates the sensitivity of the choice to the stimulus and is usually measured by the slope at the center point or the steepest slope. The *upper lapse rate* κ captures the probability of an incorrect response even at high stimulus levels, and the *lower lapse rate* γ captures stimulus-independent "errors" in the opposite tail.

A fitted curve describes behavior in one environment, but it does not specify how its parameters should move when the environment changes, hence lacking power on how choices shift. The model under Fisher information cost provides these comparative-static predictions. We organize its predictions by what changes: the incentives (Section 3.3.1) and the distribution of stimuli (Section 3.3.2). Section 3.3.3 collects the predictions in psychometric terms and turns them into tests.

3.3.1 Changes in incentives

Proposition 8 (Stronger incentive sharpens the response). *Suppose the prior satisfies Assumption 2 and is symmetric. Let u_1, u_2 satisfy Assumption 1 and be antisymmetric about c, and p_1^*, p_2^* be the corresponding optimal responses. Suppose the incentive is larger at every state under u_2, i.e. $u_2(\theta) \geq u_1(\theta) > 0$ on $(c, b]$. Then,*

$$\left|p_2^*(\theta) - \tfrac{1}{2}\right| \geq \left|p_1^*(\theta) - \tfrac{1}{2}\right| \text{ for every } \theta \in \Theta, \qquad (p_2^*)'(c) \geq (p_1^*)'(c) > 0.$$

Moreover, if $u_2(\theta) > u_1(\theta)$ *on a set of positive measure, then* $\left|p_2^*(\theta) - \frac{1}{2}\right| > \left|p_1^*(\theta) - \frac{1}{2}\right|$ *for every* $\theta \neq c$, *and* $(p_2^*)'(c) > (p_1^*)'(c) > 0$.

Raising the incentives symmetrically reduces mistakes at every state and steepens the curve at the threshold. In psychometric terms, sensitivity rises and both lapse rates fall, while the threshold stays at c.

How fast do mistakes disappear as stakes scale up? Proposition 8 establishes the qualitative prediction that stronger incentives improve accuracy in a symmetric environment. We now quantify how rapidly the probability of making a mistake declines as incentives increase.

Consider a *fixed-prize discrimination task* with a uniform prior on $[0, 1]$. The decision maker receives a reward $U > 0$ for choosing the correct action and zero otherwise. Thus, the relative payoff is $u = -U$ on $[a, c)$ and $u = U$ on $(c, b]$. The probability of making a mistake at θ and the average error rate are respectively:

$$e_U(\theta) = \begin{cases} p_U^*(\theta), & \theta < c, \\ 1 - p_U^*(\theta), & \theta > c, \end{cases} \qquad E_U = \int_a^b g(\theta)\, e_U(\theta)\, d\theta$$

Proposition 9 (Diminishing returns to incentives). *In the baseline discrimination task, as* $U \to \infty$*:*

(i) the average error rate falls at the rate $1/\sqrt{U}$*:*

$$\lim_{U\to\infty} \sqrt{\frac{U}{\lambda}}\, E_U = \frac{4 - 2\sqrt{2}}{b - a};$$

(ii) the error at each fixed state $\theta \neq c$ *falls exponentially in* $\sqrt{U}$*:*

$$\lim_{U\to\infty} \frac{1}{\sqrt{U}} \log e_U(\theta) = -\frac{|\theta - c|}{\sqrt{\lambda}}.$$

At high stakes, the average error rate decreases at rate $1/\sqrt{U}$, so quadrupling the reward only halves the average error. At any fixed state θ, mistakes vanish rapidly when the stakes are high, of order $\exp(-|\theta - c|\sqrt{U/\lambda})$. These together predict the following picture: stronger incentives make the response increasingly resemble the fully informed cutoff rule; Mistakes become rare at states far from the decision boundary and concentrate in an increasingly narrow region around the cutoff. The width of this region shrinks at the rate $1/\sqrt{U}$, which determines the rate of decline in the average error probability.

Under a linear mutual information cost, equation (1) shows that the probability of incorrect choice remains constant across ever state and vanish exponentially fast with the incentive. For comparison, we calibrate both models so that the average error equals 20 percent at a baseline reward U_0, and hold the cost parameters fixed. Table 2 reports the implied average error at

moderate levels of stakes. Doubling the reward lowers the average error to 12.9 percent under the Fisher information cost and to 5.9 percent under mutual information. At eight times the baseline reward, the two predictions are 6.0 percent and 0.0015 percent. The models disagree substantially already for moderate changes in stakes, so the prediction can be tested without extrapolating to extreme rewards. Figure 4 plots the comparison and the error profiles behind it.

Reward relative to baseline, U/U_0	0.5	1	2	4	8
Fisher information cost	29.6	20.0	12.9	8.6	6.0
Linear mutual information	33.3	20.0	5.9	0.39	0.0015

Table 2: Average error rate in the fixed-prize discrimination task with uniform prior. Both cost parameters are calibrated so that $E_{U_0} = 0.2$.

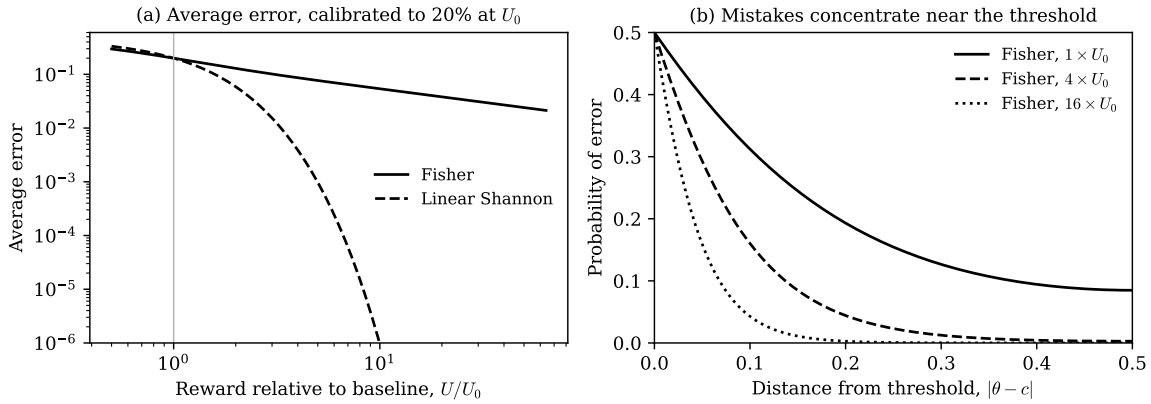


Figure 4: Incentives and mistakes in the fixed-prize discrimination task.

Asymmetric incentives. Next, we consider changes in incentives that are not symmetric.

Proposition 10 (Rewarding an action raises its choice probability everywhere). *Fix a prior g and a scaling parameter $\lambda > 0$. Let u_L, u_H be two payoffs and p_L^*, p_H^* be the corresponding optimal responses. If $u_H(\theta) \geq u_L(\theta)$ for all θ, Then,*

$$p_H^*(\theta) \geq p_L^*(\theta), \qquad \text{for every } \theta \in [a, b].$$

If the payoffs differ on a set of positive measure, and at least one optimal response is interior, then the inequality is strict at every state.

An local increase in the payoff of action R raises its choice probability everywhere, including states whose payoffs are unchanged. This result does not require single-crossing payoffs and holds under any strictly positive prior. The mutual information benchmark delivers a similar qualitative prediction. Raising payoff anywhere increases the unconditional frequency

q in (1), therefore the probability of choosing R shifts up at every state. In psychometric terms, the 50-percent point moves toward the states that favor L, the lower lapse $p^*(a)$ rises, while the upper lapse $1 - p^*(b)$ falls.

3.3.2 Changes in prior beliefs

Proposition 11 (Prior concentration flattens the response). *Under Assumptions 1–3, suppose the payoff is antisymmetric and single-crossing. Let g_1 and g_2 be symmetric priors, and let p_1^* and p_2^* denote the corresponding optimal responses. Suppose g_2 is more concentrated around the midpoint c than g_1 in the central monotone likelihood-ratio sense, $g_2(\theta)/g_1(\theta)$ is weakly decreasing in $|\theta - c|$. Then*

$$\left|p_2^*(\theta) - \tfrac{1}{2}\right| \le \left|p_1^*(\theta) - \tfrac{1}{2}\right| \quad \text{for every } \theta, \qquad 0 < (p_2^*)'(c) \le (p_1^*)'(c).$$

Moreover, if g_2/g_1 is not constant, then $\left|p_2^*(\theta) - \frac{1}{2}\right| < \left|p_1^*(\theta) - \frac{1}{2}\right|$ for every $\theta \neq c$, and $0 < (p_2^*)'(c) < (p_1^*)'(c)$.

Concentrating stimuli near the threshold flattens the response curve and raises errors at every stimulus, including physically identical ones. This cannot be explained by hard cases being sampled more often, because errors rise at each fixed stimulus. In a fixed-prize task, the payoff available beyond the threshold, $\int_c^b gu = U/2$, is the same under every symmetric prior, while the cost of steepness at the threshold is proportional to $g(c)$; for instance, the residual incentive ratio there is $\mathcal{R}(c) = U/(2g(c))$. Concentrating the prior raises $g(c)$ and leaves the benefit unchanged, so the curve flattens. Intuitively, when hard cases are common, discriminating them finely is expensive, because the cost of attention is paid on every trial.

By contrast, under mutual information, the response is unchanged as long as the prior probability on each side of the threshold is fixed. As shown in (1), the prior enters only through the unconditional probability q, which equals $1/2$ under every symmetric prior and antisymmetric payoff.

Proposition 12 (Increased base rate shifts choice everywhere). *Fix a strictly single-crossing payoff u and a scaling parameter $\lambda > 0$. Let g_L, g_H be two strictly positive C^1 priors and p_L^*, p_H^* be the corresponding optimal responses. If g_H puts more weight on higher states than g_L in the monotone likelihood ratio sense: $g_H(\theta)/g_L(\theta)$ is nondecreasing in θ. Then,*

$$p_H^*(\theta) \ge p_L^*(\theta), \quad \text{for every } \theta \in [a, b].$$

If the likelihood ratio is nonconstant, and at least one optimal response is interior, then the inequality is strict for every $\theta \in [a, b]$.

A prior that places more weight on the states where R is favored raises the probability of choosing R everywhere. The result requires neither monotone payoffs nor a symmetric prior. The prior enters the optimality condition through its score g'/g, and a likelihood-ratio shift toward higher states acts on the response like an increase in the relative payoff. The

psychometric implications is also consistent with the Bayesian intuition: when states favoring R are more prevalent, the decision maker requires less evidence to choose R, shifting the psychometric curve upward. For instance, a heightened prior presumption of guilt leads a judge to convict more frequently across every level of evidence.

3.3.3 Mapping predictions into standard psychometric fits

The implication also connects with how psychophysics typically summarizes perception data. A standard specification is

$$p(\theta) = \gamma + (1 - \gamma - \kappa)F\left(\frac{t - \alpha}{\beta}\right),$$

where F is the CDF of a Gaussian or logistic distribution. The result shows that scaling up incentives decreases the lapse rate and increases sensitivity.

To illustrate how this prediction would translate into the parameters commonly reported in psychophysics, we conduct a simulation exercise in which the predicted optimal response curve is treated as the data-generating process. We consider the two-alternative forced-choice environment, with a uniform prior and a symmetric step payoff

$$u(t) = \begin{cases} -U, & t \in [0, \frac{1}{2}), \\ U, & t \in (\frac{1}{2}, 1], \end{cases}$$

where $(U > 0)$ indexes the incentive for a correct response. For each value of $U \in \{20, 40, 80, 160\}$, we solve for the optimal response rule $(p_U^*(t))$. We then generate binomial responses-count data using $(p_U^*(t))$ as the success probability with $N = 100$ trials at a grid of 19 stimulus levels. On the grid of stimulus levels, we fit by maximum likelihood the logistic specification

$$\widehat{p}(\theta) = \gamma + (1 - \gamma - \kappa)\frac{1}{1 + \exp\left(-(t - \alpha)/\beta\right)}.$$

Here (α) is the threshold parameter, (β) is the scale parameter, so that lower (β) corresponds to higher sensitivity, and (γ) and (κ) are the lower and upper lapse rates. The fitted parameters and Root Mean Squared Error (RMSE) with 90% bootstrap confidence interval are reported in Table 3.

Table 3: Bootstrap Estimates of Psychometric Parameters under Scaled Incentives

	Incentive Level (U)			
Parameter	$U = 20$	$U = 40$	$U = 80$	$U = 160$
Sensitivity ($1/\beta$)	7.807	9.050	11.598	15.842
	[4.323, 11.779]	[6.957, 11.542]	[9.802, 13.688]	[13.683, 18.305]
Threshold (α)	0.498	0.500	0.500	0.500
	[0.436, 0.558]	[0.473, 0.528]	[0.484, 0.517]	[0.489, 0.511]
Lower Lapse (γ)	0.208	0.097	0.031	0.009
	[0.105, 0.275]	[0.048, 0.139]	[0.005, 0.054]	[0.000, 0.021]
Upper Lapse (κ)	0.209	0.096	0.031	0.009
	[0.107, 0.275]	[0.044, 0.140]	[0.008, 0.055]	[0.000, 0.022]
RMSE	0.040	0.035	0.029	0.022
	[0.029, 0.054]	[0.025, 0.047]	[0.020, 0.040]	[0.014, 0.032]

Notes: Point estimates represent the mean across 1,000 bootstrap iterations. 90% confidence intervals [5th, 95th percentiles] are reported in brackets below each estimate. Simulated data consists of 100 binary choices generated at 19 uniformly spaced interior stimulus levels ($t \in [0.05, 0.95]$) for each condition. The true response probabilities were derived numerically from the Fisher information boundary-value problem.

The finite-sample exercise shows how the model's structural comparative statics would be recovered from the type of choice-frequency data observed in a laboratory psychophysics experiment. As incentives increase, the fitted sensitivity measure ($1/\beta$) increases and both lapse-rate estimates decrease while the threshold remains approximately fixed in the symmetric environment. Figure 5 illustrates these comparative statics and shows that the fitted logistic curves closely approximate the structural Fisher curves.

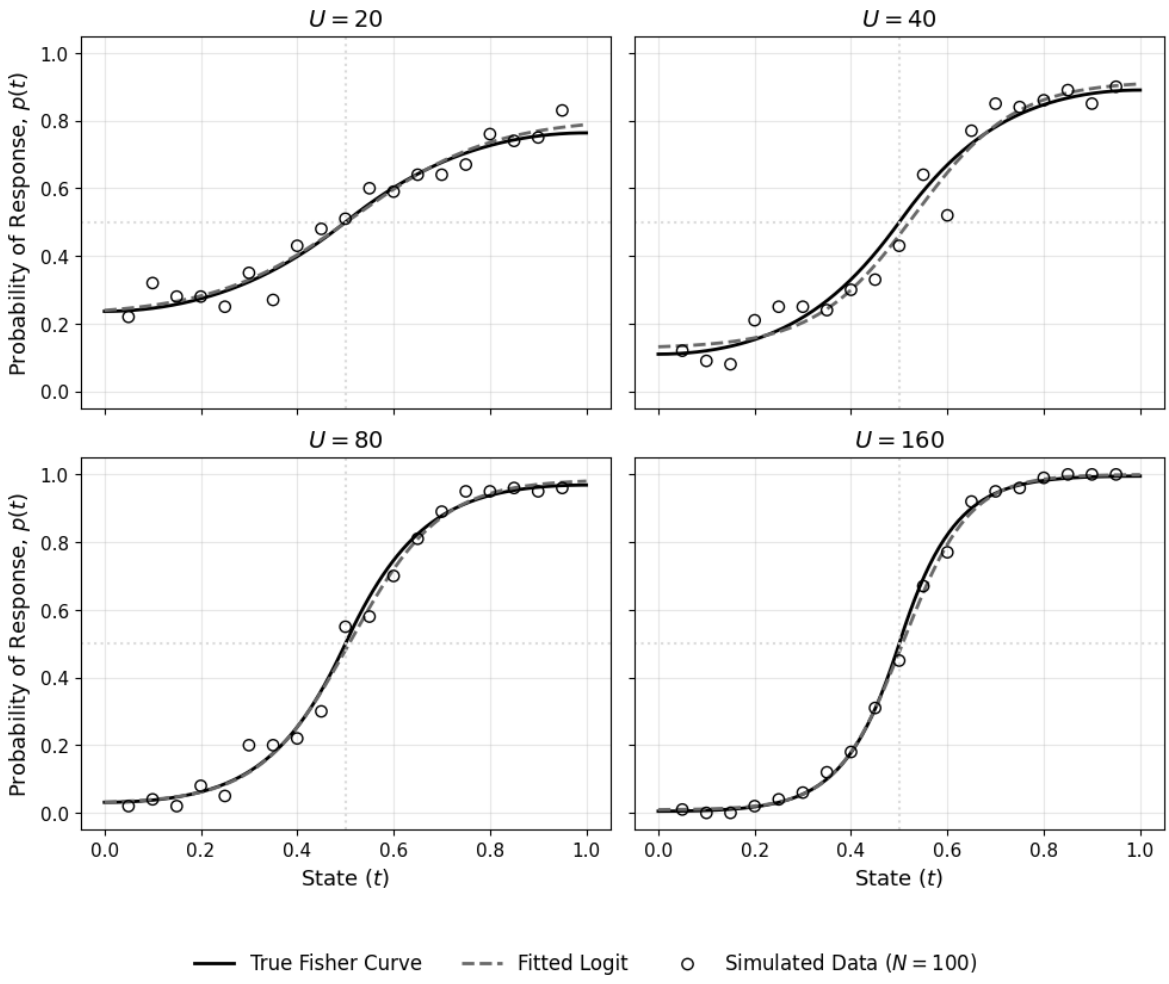


Figure 5: Optimal Response Curves and Simulated Experimental Fits

4 Characterizing the Optimal Response

This section provides the formal results underlying our behavioral implications in Section 3. We first introduce an angular reparameterization of the choice probability, under which the Fisher information cost becomes quadratic in the slope. In this angular coordinate, we derive the necessary condition for an optimal response, which is a simple boundary value equation. The equation has a natural mechanical interpretation and directly yields the smoothness properties used in Section 3.2. We then establish the existence of the optimal response and show that an optimum is either a pure constant response or a strictly interior one. The concavity of the problem ensures that any nonconstant optimum is unique. Finally, our marginal-value-of-attention criterion determines whether the optimum is interior or one of the two pure responses, providing foundations for the participation results in Section 3.1. Together, these results fully characterize the optimal response.

4.1 Geometric reparametrization

A stochastic choice rule $\theta \mapsto (p_1(\theta), \ldots, p_N(\theta))$ can be viewed as a curve in the probability simplex, parametrized by the state. Fisher information $\mathcal{I}_F(\theta) = \sum_i (p_i(\theta)')^2/p_i(\theta)$ measures how infinitesimal changes in the state lead to changes in choice probabilities. It hence defines a notion of infinitesimal distance, the so-called Fisher-Rao metric, with squared length:[10]

$$ds^2 = \sum_i^N \frac{(dp_i)^2}{p_i} = \sum_i^N \frac{(p_i'(\theta))^2}{p_i(\theta)} d\theta^2$$

The Fisher information measures the squared speed $(ds/d\theta)^2$ of the curve and the Fisher information cost $\int g\mathcal{I}_F \, d\theta$ corresponds to the (prior-weighted) kinetic energy. It captures not just how far the choice probabilities change but also how rapidly they change across states.

The weights $1/p_i$ in the metric can become unbounded as a choice probability approaches zero. It is standard to apply the square-root transformation $\phi_i = \sqrt{p_i}$ (Amari and Nagaoka, 2000, Section 2.3). Since $\sum_{i=1}^N p_i = 1 \Leftrightarrow \sum_{i=1}^N \phi_i^2 = 1$, this transformation maps the probability simplex into the positive orthant of the unit sphere. Moreover, because

$$\sum_{i=1}^N \frac{(p_i'(\theta))^2}{p_i(\theta)} = 4\sum_{i=1}^N (\phi_i'(\theta))^2,$$

it removes the state-dependent weight $1/p_i$ from the quadratic term and allows us to analyze the information cost with the more familiar geometry of the sphere.

In the binary case, the square-root image is the positive quarter of a unit circle and it is more convenient to work with the rescaled and centered angular coordinate w. More specifically, we consider the following change of variable: Given $p \in AC(\Theta, [0,1])$, we define w as the unique solution to $p(\theta) = \frac{1+\sin(w(\theta))}{2}$ for each $\theta \in \Theta$, in other words, $w(\theta) = \arcsin(2p(\theta) - 1)$.[11] Our problem becomes:

$$\underset{w \in AC(\Theta, [-\frac{\pi}{2}, \frac{\pi}{2}])}{\text{maximize}} \int_\Theta g(\theta) \left[u(\theta) \frac{1 + \sin(w(\theta))}{2} - \lambda (w'(\theta))^2 \right] d\theta. \tag{W}$$

Before working with Problem (W) in place of Problem (P), we need to verify that this change of variable does not exclude any candidate optimal choice rule. Although the transformation is pointwise bijective, the map $p \mapsto \arcsin(2p-1)$ has infinite slope at $p \in \{0,1\}$, so absolute continuity of p does not by itself imply absolute continuity of w.

However, absolute continuity of w is recovered once we restrict attention to choice rules with finite Fisher information cost. This restriction is without loss of generality: because the payoff u is bounded and uninformative constant rules incur zero cost, any optimal solution to

[10] Formally, the Fisher-Rao metric is the Riemannian metric on the simplex whose quadratic form is the Fisher information. The induced Fisher-Rao distance quantifies the statistical distinguishability between two probability distributions.

[11] Let $\sqrt{p} = \sin\phi$ with $\phi \in [0, \pi/2]$ and set $w = 2\phi - \pi/2$, we get $p = (1 + \sin w)/2$. Consequently, $p'(\theta) = \cos(w(\theta))w'(\theta)/2$ and $p(\theta)(1-p(\theta)) = (1 - \sin^2(w(\theta)))/4$.

Problem (P) must have finite cost. It is therefore equivalent to optimize over Problem (W).

Lemma 4.1 (Bijection). *The change of variables*

$$w = \arcsin(2p-1) \iff p = \frac{1+\sin w}{2}$$

induces a bijection between the sets of feasible choice rules with finite cost in Problems (P) and (W), and preserves the objective value. Consequently, p^* solves Problem (P) if and only if $w^* = \arcsin(2p^*-1)$ solves Problem (W).

Proof. See Appendix B.2. □

4.2 Optimality equation

To describe the optimal solution, we first derive the necessary first-order condition. The change of variable above makes this first-order condition particularly simple:

Theorem 1 (Necessity). *Under Assumptions 1-2, let w^* be a solution to Problem (W), then w^* satisfies:*

$$\begin{cases} g(\theta)(w^*)'(\theta) = -\frac{1}{4\lambda}\int_a^\theta g(s)u(s)\cos(w^*(s))ds \quad \text{a.e. on } [a,b] \\ (w^*)'(a) = (w^*)'(b) = 0 \end{cases} \tag{NC}$$

Proof. See Appendix B.4. □

Remark. We might also write the integral equation as a second-order ordinary differential equation, in the sense of **Carathéodory**[12] (Walter, 1998, §10):

$$w''(\theta) + \frac{g'(\theta)}{g(\theta)}w'(\theta) = -\frac{1}{4\lambda}u(\theta)\cos(w(\theta)) \quad \text{for almost all } \theta.$$

4.2.1 A mechanical interpretation

Under uniform prior, the optimality equation reduces to

$$\begin{cases} w''(\theta) = -\frac{1}{4\lambda}u(\theta)\cos(w(\theta)) \text{a.e. on } [a,b] \\ w'(a) = w'(b) = 0 \end{cases}$$

It admits a physical interpretation as a unit-mass object attached to a pivot by a rigid, unit-length rod, and subject to a time-varying external force acting in the horizontal direction.[13] The interval $[a,b]$ represents time, with $w(\theta)$ denoting the angle between the rod and the vertical axis at time θ, and $\tilde{u}(\theta) = \frac{1}{4\lambda}u(\theta)$ representing the applied force. Angles to the left of

[12] A function y is a solution to a differential equation in the sense of **Carathéodory** if y is absolutely continuous and satisfies the differential equation almost everywhere.

[13] We assume there is no gravitational force acting on the object.

the vertical midline are taken to be negative, and horizontal forces directed to the right are taken to be negative.

A standard force decomposition and Newton's second law imply that the object experiences a tangential acceleration of $\tilde{u}(\theta)\cos(w(\theta))$ along its circular trajectory. The boundary conditions $w'(a) = w'(b) = 0$ then translate into the requirement that the object starts and ends at rest, i.e., with zero angular velocity at both endpoints. Figure 6 illustrates the setup.

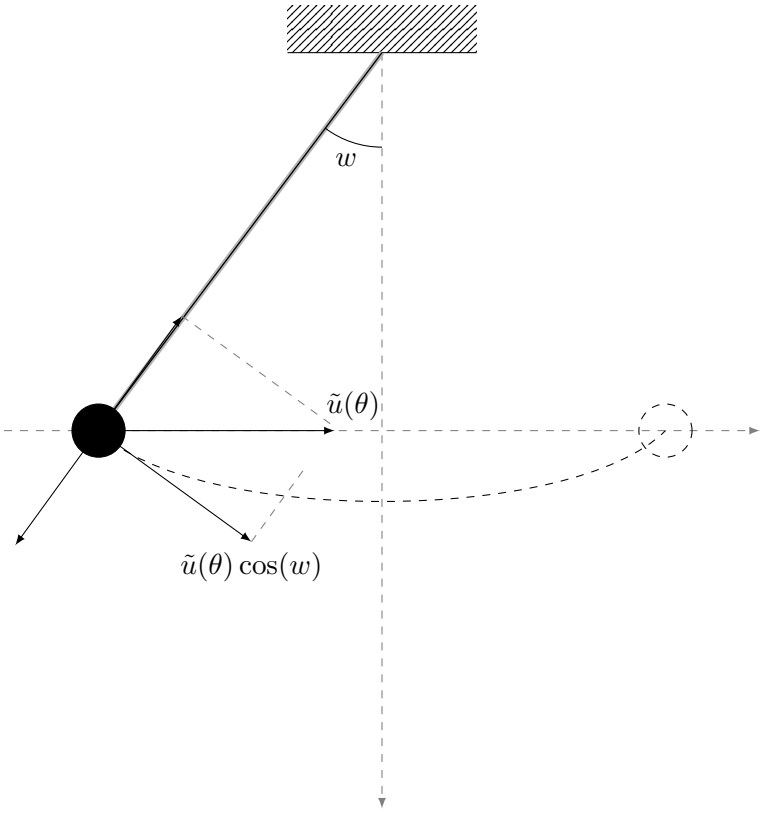


Figure 6: A pendulum with varying force

4.2.2 Regularity from Euler equation

The optimality equation says that the acceleration w'' is a bounded function of the state, the position w and the velocity w'. Integrating once gives a continuous velocity and integrating twice a continuously differentiable path, whatever discontinuities the force u may have. In the pendulum, a sudden change in the force changes the acceleration at once, but position and velocity move on continuously. Translated back, the response curve p^* has no jumps and no kinks: a discontinuity in incentives, such as the fixed prize for a correct answer that switches sign at the threshold, shows up only in the curvature of the response, never in its level or slope. More generally, the response is two derivatives smoother than the primitives, and the only feature of the prior that matters for this is its score g'/g, which enters the equation as a velocity-dependent force.

Theorem 2 (Regularity). *Let p^* be the solution to Problem (P). If u is piecewise continuous*

and g is absolutely continuous, then p^* is twice differentiable almost everywhere. Furthermore, if $u \in C^k$ and $g \in C^{k+1}$ for some $k \in \mathbb{N}$, then $p^* \in C^{k+2}$.

Proof. See Appendix B.7. □

4.3 Existence and uniqueness

4.3.1 Existence of the optimal solution

The angular reparameterization makes existence of solution transparent: the cost is a positive weighted quadratic form in w', and w lies in a compact interval. A maximizing sequence is therefore bounded in H1, and the direct method yields an optimum. The optimality equation then implies that the optimum is Lipschitz continuous.[14]

We might also care whether our constraint $0 \leq p \leq 1$ is binding. It turns out that, for the optimal solution, the constraint is either always binding or not binding at all, but it cannot be active on some intervals and inactive on other intervals. That is, under the optimal signal structure, the agent either learns nothing about the state and sticks to one action or she never chooses an action with certainty under whichever state. This helps us to simplify the necessary condition, so we do not need to impose an extra multiplier for the constraint.

The following statement summarizes the result, and the proof can be found in the appendix:

Theorem 3 (Existence). *Under Assumptions 1-2, Problem (P) admits a solution in Lip$([a,b],[0,1])$. Furthermore, every optimal solution is to one of the following:*

- *Pure boundary solution: $p(\theta) = 0$ for all $\theta \in [a,b]$ or $p(\theta) = 1$ for all $\theta \in [a,b]$*
- *Strict interior solution: $0 < p(\theta) < 1$ for all $t \in [a,b]$*

Proof. See Appendix B.5 □

4.3.2 Uniqueness of optimum

We now study the uniqueness property of the solution to our problem. First, let $\mathcal{D}^P$ denote the admissible choice rules with finite Fisher information cost, and denote

$$J^P[p] := \int_a^b g(\theta) \left[u(\theta)p(\theta) - \lambda \frac{(p'(\theta))^2}{p(\theta)(1-p(\theta))} \right] d\theta$$

We note the integrand of the maximization Problem (P) is jointly concave in (p, p').[15]

[14] A function $f : [a,b] \to \mathbb{R}$ is said to be Lipschitz continuous if there exists a constant $K > 0$ such that

$$|f(t_1) - f(t_2)| \leq K|t_1 - t_2| \quad \text{for any } t_1, t_2 \in [a,b].$$

[15] The integrand of the transformed problem (W) is not concave in (w, w'), and this is not contradictory. Consider two candidates w_1, w_2 and their corresponding transformation $p_1 = \frac{1+\sin(w_1)}{2}, p_2 = \frac{1+\sin(w_2)}{2}$, the transformation of the convex combination is in general not the convex combination of the transformations, i.e. $\frac{1+\sin(\lambda w_1 + (1-\lambda)w_2)}{2} \neq \lambda \frac{1+\sin(w_1)}{2} + (1-\lambda)\frac{1+\sin(w_2)}{2}$

Lemma 4.2 (Concavity). *The set $\mathcal{D}^P$ is convex, and J^P is concave on $\mathcal{D}^P$. For any $p_1, p_2 \in \mathcal{D}^P$ and $\alpha \in (0,1)$,*

$$J^P[\alpha p_1 + (1-\alpha)p_2] \geq \alpha J^P[p_1] + (1-\alpha)J^P[p_2]$$

The equality holds if and only if $p_1 = p_2$ or both p_1 and p_2 are constant.

Proof. See Appendix B.3. □

Under Assumption 3 that u is nonzero on some set of positive measure, we further know that the optimal choice rule is unique. This is because, suppose two optimal choice rule exist, their mixture is also optimal, and the lemma above implies that both choice rules must be constant rules. However, an interior constant rule $p = c \in (0,1)$ cannot be optimal. To see this, consider the perturbation $c + \epsilon h$ for some smooth h, the first variation is:

$$\frac{d}{d\epsilon} J^P[c + \epsilon h]\Big|_{\epsilon=0} = \int_a^b g(\theta)u(\theta)h(\theta)d\theta,$$

optimality requires this integral to be zero for every smooth h, forcing $u = 0$ almost everywhere and contradicting Assumption 3.

Corollary 3 (Uniqueness). *Under Assumption 3, Problem (P) has a unique solution.*

4.4 Boundary versus interior optimum

The optimality equation alone does not characterize the optimal response, as it is always satisfied by $w^L \equiv -\frac{\pi}{2}$ and $w^R \equiv \frac{\pi}{2}$, which correspond to the choice rule of always choosing L and always choosing R.

We determine whether either of the constant rules is optimal by checking whether a small departure is profitable. To check whether always choosing L is optimal, we inspect small deviations from this choice rule. Specifically, we consider perturbation profiles that have equal weighted L^2 norm, that is, all $h \in AC([a,b])$ satisfying $\int gh^2 = 1$. Two immediate remarks follow: since h is continuous and hence bounded, $w^L_{\epsilon,h}$ falls in $[-\pi/2, \pi/2]$ for all sufficiently small ϵ, and we can drop the sign restriction of this one-sided deviation because $\sin(-\pi/2 + \epsilon|h|) = \sin(-\pi/2 + \epsilon h)$ and $(|h|')^2 = (h')^2$ almost everywhere. The normalization also fixes the scale of the perturbation in probability coordinate: $\int_a^b g p_{\epsilon,h} = \epsilon^2/4 + o(\epsilon^2)$.

Write J^W as the objective in Problem (W), and let the deviating rule be

$$w^L_{\epsilon,h}(\theta) := -\tfrac{\pi}{2} + \epsilon h(\theta).$$

The deviation leads to a change in the net expected payoff:

$$J^W[w^L_{\epsilon,h}] - J^w[w^L] = \frac{\epsilon^2}{4}\Big[\int guh^2 - 4\lambda \int g(h')^2\Big] + o(\epsilon^2)$$

Let $Q_L[h] := \int_a^b guh^2 - 4\lambda \int_a^b g(h')^2$ denote the marginal net benefit of deviation from choosing L via a perturbation h. We call the marginal benefit of the best such deviation as the *marginal value of attention* at always-L:

$$m_L := \sup\{Q_L[h] : h \in AC([a,b]), \int_a^b gh^2 = 1\}.$$

The case of deviating from always-R is similar, with the deviating rule $w^R_{\epsilon,h}(\theta) := \frac{\pi}{2} + \epsilon h(\theta)$,

$$Q_R[h] := -\int guh^2 - 4\lambda \int g(h')^2, \quad J[w^R_{\epsilon,h}] - J[w^R] = \frac{\epsilon^2}{4} Q_R[h] + o(\epsilon^2),$$

and the *marginal value of attention* at always-R is:

$$m_R := \sup\{Q_R[h] : h \in AC([a,b]), \int_a^b gh^2 = 1\}.$$

If $m_L > 0$, then there exists some small profitable deviation from always-L, so it cannot be optimal. It turns out that, the converse is also true, that is, if no small deviation is profitable, then no deviation of any size is:

Theorem 4 (Interiority). *Exactly one of the following three cases holds:*

(i) the marginal value of attention m_L is non-positive, and always-L is the unique optimal response;

(ii) the marginal value of attention m_R is non-positive, and always-R is the unique optimal response;

(iii) both m_L and m_R are strictly positive, and the optimal response w^* is strictly interior, i.e. $-\pi/2 < w^*(\theta) < \pi/2$ for each $\theta \in [a,b]$

Proof. See Appendix B.6 □

In general, the check of profitable infinitesimal deviation only provides a local test. To see why a local test also decides the global optimality, notice that Problem (P) is jointly concave in (p, p'), which turns a local maximum into a global maximum. Alternatively, Appendix B.6 shows that the quadratic form above is not merely an approximation, it also gives a global upper bound of deviation benefit. This also explains why the knife-edge case of $m_L = 0$ or $m_R = 0$ also lead to a corner solution: when the best small deviation cannot bring a strict benefit, scaling it up cannot either.

Remark (Relation to Hébert and Woodford (2021))**.** Parts (i) and (ii) of Theorem 4 coincide with conditions (15)–(16) of their Proposition 3, under the reformulation $\theta_{HW} = 4\lambda$ and $p_L = h^2$. Our formulation emphasizes the marginal deviation interpretation and the criterion is operational: m_L and m_R can be computed from a linear variational problem and bounded using simple trial profiles. These properties help to derive the behavioral implications in Section 3.1.

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A Proof of results in Section 3

A.1 Proof of Proposition 1 and Proposition 2 (Marginal value of attention)

Proof of Proposition 1. If $\int gu = 0$ and $u \not\equiv 0$, we can choose a smooth h with $A := \int guh > 0$. Such a test function exists because the signed density gu is nonzero, and we can take h to be a smooth approximation of gu (since smooth functions are dense in L^2 and $gu \in L^2$), ensuring that $\int guh$ is arbitrarily close to $\int (gu)^2 > 0$.

For sufficiently small $\varepsilon > 0$, $p_\varepsilon = 1/2 + \varepsilon h$ is feasible and

$$\begin{aligned} J^P[p_\varepsilon] &= \varepsilon A - 4\lambda\varepsilon^2 \int_a^b \frac{g(\theta)h'(\theta)^2}{1 - 4\epsilon^2 h(\theta)^2} d\theta \\ &= \varepsilon A - O(\varepsilon^2) > 0. \end{aligned}$$

Since every constant rule has value zero, paying attention is optimal for every finite $\lambda > 0$. □

Proof of Proposition 2. If both signs of payoff occur with positive measure, we can choose a nonzero smooth bump h supported where $u > 0$ with $\int guh^2 > 0$. Piecewise continuity of u ensures suitable intervals and hence such h exist.

We have $Q_L[h] = \int guh^2 - 4\lambda \int gh'^2 > 0$ for sufficiently small λ. By Theorem 4, always choosing-L is not optimal for small enough λ.

A bump where $u < 0$ similarly gives $Q_R > 0$ for sufficiently small λ. Since both pure boundary choice rule are not optimal, the optimal response is interior and the incurred information cost is strictly positive. □

A.2 Proof of Proposition 4 (Monotonicity)

Proof. Suppose the solution is a constant, then it is certainly weakly increasing. Otherwise, if it is a strict interior solution, then the corresponding solution w_* to Problem (W) must satisfy the integral Euler equation in (NC). In this case, $w^*(\theta) \in (-\frac{\pi}{2}, \frac{\pi}{2}) \Rightarrow \cos(w_*(\theta)) > 0$ and

$$g(\theta)(w^*)'(\theta) = -\frac{1}{4\lambda} \int_a^\theta g(s)u(s)\cos(w_*(s))ds = \frac{1}{4\lambda} \int_\theta^b g(s)u(s)\cos(w_*(s))ds,$$

hence $g \cdot (w^*)'$ is strictly increasing when $u(s) < 0 \Rightarrow g(s)u(s)\cos(w_*(s))$ and strictly decreasing when $u(t) > 0$. Combining with the boundary condition $(w^*)'(a) = (w^*)'(b) = 0$, we know $(w^*)'(\theta) > 0$ for all $\theta \in (a, b)$.

Since $p^* = (1 + \sin(w^*))/2$, we know $(p^*)' = \cos(w_*)w'_* \geq 0$ on $\Theta = [a, b]$ and $(p^*)' > 0$ on (a, b) if p^* is nonconstant. □

A.3 Proof of Proposition 5 (S-shape in symmetric tasks)

Proof. By Proposition 1, the interior solution w^* exists for every $\lambda > 0$. Then, by Proposition 4, the solution w^* is strictly increasing. Now, let $c = (a+b)/2$ and consider the reflection $\hat{w}(\theta) = -w^*(2c-\theta)$. Its derivatives satisfy:

$$(\hat{w})'(\theta) = (w^*)'(2c-\theta) \text{ and } (\hat{w})''(\theta) = -(w^*)''(2c-\theta),$$

Substituting it into the Optimality equation:

$$\begin{aligned} g(\theta)(\hat{w})'(\theta) = g(2c-\theta)(w^*)'(2c-\theta) &= -\frac{1}{4\lambda}\int_a^{b-\theta} g(s)u(s)\cos(w^*(s))\,ds \\ &= -\frac{1}{4\lambda}\int_b^{a+\theta} g(2c-s)u(2c-s)\cos(w^*(2c-s))\,d(2c-s) \\ &= -\frac{1}{4\lambda}\int_a^{\theta} g(s)u(s)\cos(\hat{w}(s))\,ds \end{aligned}$$

we know $\hat{w}$ is another solution to the Optimality equation. Then, by the uniqueness of solution from Corollary 3, w^* must be symmetric around c:

$$w^*(\theta) = \hat{w}(\theta) = -w^*(2c-\theta).$$

Sufficiency for S-shaped: From the Optimality equation,

$$\begin{aligned} (w^*)'(\theta) &= \frac{\int_\theta^b g(s)u(s)\cos(w(s))ds}{4\lambda g(\theta)} \\ &= \frac{1}{4\lambda}\frac{\int_\theta^b g(s)u(s)\cos(w(s))ds}{\int_\theta^b g(s)u(s)ds}\frac{\int_\theta^b g(s)u(s)ds}{g(\theta)} \end{aligned}$$

We first note that $H(\theta) := \frac{\int_\theta^b g(s)u(s)\cos(w(s))ds}{\int_\theta^b g(s)u(s)ds}$ is decreasing on $[c,b]$. Because on $(c,b]$, $g>0$, $u \geq 0$ and $w^* \in (0,\pi/2)$ is strictly increasing , we know $\cos(w^*)$ is strictly decreasing, so

$$\begin{aligned} H(\theta) &= \frac{\int_\theta^b g(s)u(s)\cos(w^*(s))ds}{\int_\theta^b g(s)u(s)ds} \leq \cos(w^*(\theta)) \\ \Rightarrow \frac{d}{d\theta}H(\theta) &= \frac{g(\theta)u(\theta)}{\int_\theta^b g(s)u(s)ds}\left[H(\theta) - \cos(w^*(\theta))\right] \leq 0 \text{ a.e.} \end{aligned}$$

Hence, if $\mathcal{R}(\theta) = \frac{\int_\theta^b g(s)u(s)ds}{g(\theta)}$ is nonincreasing, $(w^*)'(\theta) = \frac{1}{4\lambda}H(\theta)\mathcal{R}(\theta)$ is also nonincreasing on $[c,b]$. Finally, $(p^*)'(\theta) = \frac{1}{2}\cos(w^*(\theta))(w^*)'(\theta) = \frac{1}{8\lambda}\cos(w^*(\theta))H(\theta)\mathcal{R}(\theta)$ is decreasing because $\cos(w^*)$ is strictly decreasing on $[c,b]$. Reflection symmetry implies that $(p^*)'(\theta)$ is increasing on $[a,c]$. Combining with Proposition 4 and Proposition 3, the solution p^* is S-shaped by Definition 3.1.

Necessity for S-shaped for all $\lambda > 0$: Denote the solution to Problem (W) under parameter

λ as w^*_λ. Since $\cos(w^*_\lambda(s)) \le 1$, $H_\lambda(\theta) := \frac{\int_\theta^b g(s)u(s)\cos(w_\lambda(s))ds}{\int_\theta^b g(s)u(s)ds} \le 1$, we know w^* shrinks uniformly to 0 as λ grows large:

$$(w^*_\lambda)'(\theta) = \frac{1}{4\lambda} H_\lambda(\theta)\mathcal{R}(\theta) \le \frac{1}{4\lambda}\mathcal{R}(\theta)$$

By reflection symmetry, $w^*_\lambda(c) = 0$. Integrating from c yields

$$0 \le w^*_\lambda(\theta) = \int_c^\theta (w^*_\lambda)'(s)\, ds \le \frac{1}{4\lambda}\int_c^b R(s)\, ds \implies \|w^*_\lambda\|_\infty = O(\lambda^{-1}).$$

Because $|1 - \cos x| \le |x|$, we also have $\|1 - \cos(w^*_\lambda)\|_\infty \le \|w^*_\lambda\|_\infty = O(\lambda^{-1}) \to 0$ as $\lambda \to \infty$.

Now consider the scaled derivative $8\lambda(p^*_\lambda)'(\theta) = \frac{\cos(w^*_\lambda(\theta))}{g(\theta)} \int_\theta^b g(s)u(s)\cos(w^*_\lambda(s))\, ds$. The difference from $\mathcal{R}(\theta)$ satisfies:

$$\begin{aligned}
\left|8\lambda(p^*_\lambda)'(\theta) - \mathcal{R}(\theta)\right| &= \frac{1}{g(\theta)}\left|\cos(w^*_\lambda(\theta))\int_\theta^b g(s)u(s)\cos(w^*_\lambda(s))\, ds - \int_\theta^b g(s)u(s)\, ds\right| \\
&\le \frac{1}{g(\theta)}\left[|1-\cos(w^*_\lambda(\theta))|\int_\theta^b g(s)u(s)\, ds + \int_\theta^b g(s)u(s)|1-\cos(w^*_\lambda(s))|\, ds\right] \\
&\le 2\mathcal{R}(\theta)\|1-\cos(w^*_\lambda)\|_\infty = O(\lambda^{-1}).
\end{aligned}$$

Thus, $8\lambda(p^*_\lambda)'(\theta) \to \mathcal{R}(\theta)$ uniformly on $[c, b]$ as $\lambda \to \infty$.

If p^*_λ is S-shaped for every $\lambda > 0$, then $(p^*_\lambda)'(\theta)$ is nonincreasing on $[c, b]$ for all sufficiently large λ. Because the limit of a sequence of nonincreasing functions is nonincreasing, the uniform limit $\mathcal{R}(\theta) = \lim_{\lambda\to\infty} 8\lambda(p^*_\lambda)'(\theta)$ must also be nonincreasing on $[c, b]$. □

A.4 Proof of Proposition 6

To highlight the main idea of the argument, we first present the proof under a stronger assumption that the payoff function u is absolutely continuous. We then extend the result to the general case where we do not require continuity.

Recall the transformation: $p(\theta) = \frac{1+\sin(w(\theta))}{2}$, which implies

$$\begin{aligned}
p' &= \frac{1}{2}\cos(w)w' \\
p'' &= \frac{1}{2}\cos(w)w'' - \frac{1}{2}\sin(w)(w')^2 \\
p''' &= \frac{1}{2}(\cos(w)w''' - 3\sin(w)w'w'' - \cos(w)(w')^3)
\end{aligned}$$

Meanwhile, we make use of the E-L equation:

$$\begin{aligned}
w'' &= -\frac{1}{4}\cos(w)u(t) \\
w''' &= -\frac{1}{4}\cos(w)u'(t) + \frac{1}{4}u(t)\sin(w)w'
\end{aligned}$$

And substitute w'', w''' in the expression of p''':

$$p''' = \frac{1}{2}\cos(w)(u(t)\sin(w)w' - \frac{1}{4}u'(t)\cos(w) - (w')^3)$$

By our assumption, there exists t_u such that $u(t) < 0$ if $t \in [0, t_u)$ and $u(t) > 0$ if $t \in (t_u, 1]$. Meanwhile, by our previous result, u being strictly single-crossing implies w is strictly increasing on $[0, 1]$ and $w'(\theta) > 0$ on $(0, 1)$. We divide our discussion into 3 cases:

Case I: w crosses 0: If there also exists a t_w such that $w(\theta) < 0$ if $t \in [0, t_w)$ and $w(\theta) > 0$ if $t \in (t_w, 1]$.

On $[0, \min\{t_u, t_w\})$, we have

$$p'' = -\frac{1}{8}\overset{>0}{\cos^2}(w)\overset{<0}{u(t)} - \frac{1}{2}\overset{<0}{\sin}(w)\overset{>0}{(w')^2} > 0$$

Similarly on $(\max\{t_u, t_w\}, 1]$, we have:

$$p'' = -\frac{1}{8}\overset{>0}{\cos^2}(w)\overset{>0}{u(t)} - \frac{1}{2}\overset{>0}{\sin}(w)\overset{>0}{(w')^2} < 0$$

On $(\min\{t_u, t_w\}, \max\{t_u, t_w\})$, we know $u(t)\sin(w(\theta)) < 0$ and hence:

$$p''' = \frac{1}{2}\overbrace{\cos(w)}^{>0}(\overbrace{u(t)\sin(w)}^{<0}\overbrace{w'}^{>0} - \frac{1}{4}\overbrace{u'(t)}^{\geq 0}\overbrace{\cos(w)}^{>0} - \overbrace{(w')^3}^{>0}) < 0$$

Since $p''(\min\{t_u, t_w\}) \geq 0$ and $p''(\max\{t_u, t_w\}) \leq 0$, and from above we know p'' decreases on $(\min\{t_u, t_w\}, \max\{t_u, t_w\})$. Therefore, there exists a τ such that $p''(t) \geq 0$ if $t \in [0, \tau)$ and $p''(t) \leq 0$ if $t \in (\tau, 1]$.

Case II: w stays above 0: Suppose $w(\theta) \geq 0$ for all t. Firstly we notice that p'' is continuous almost everywhere because boundedness of u implies w' is Lipschitz-continuous and u being continuous almost everywhere implies w'' is also continuous almost everywhere. Meanwhile, since $w'(0) = 0 \Rightarrow p''(0) = -\frac{1}{8}\cos^2(w(0))u(0) > 0$, and $w'(1) = 0 \Rightarrow p''(1_-) = -\frac{1}{8}\cos^2(w(1))u(1) < 0$, we can assume

$$t_p = \sup\{s : p''(t) > 0 \text{ on } [0, s)\}$$

On $[t_u, 1]$, we have:

$$p'' = -\frac{1}{8}\overset{>0}{\cos^2}(w)\overset{>0}{u(t)} - \frac{1}{2}\overset{>0}{\sin}(w)\overset{>0}{(w')^2} < 0$$

this implies that $t_p < t_u$ On $[0, t_u)$, since $u(t) < 0$ and $\sin(w) > 0$ we again have:

$$p''' = \frac{1}{2}\overbrace{\cos(w)}^{>0}(\overbrace{u(t)\sin(w)}^{<0}\overbrace{w'}^{>0} - \frac{1}{4}\overbrace{u'(t)}^{\geq 0}\overbrace{\cos(w)}^{>0} - \overbrace{(w')^3}^{>0}) < 0$$

So p'' decreases on $[0, t_u)$. Therefore, as we have assumed, $p''(t) \geq 0$ if $t \in [0, t_p)$, and $p''(t) \leq 0$ if $t \in (t_p, 1]$.

Case III: w stays below 0: On $[0, t_u]$, we have:

$$p'' = -\frac{1}{8}\overset{>0}{\cos^2}(w)\overset{<0}{u}(t) - \frac{1}{2}\overset{<0}{\sin}(w)\overset{>0}{(w')^2} > 0,$$

Similarly, we assume

$$t_p = \sup\{s : p''(t) > 0 \text{ on } [0, s)\},$$

and we must have $t_p > t_u$.

On $(t_u, 1]$, $u(t)\sin(w) < 0$ we have:

$$p''' = \frac{1}{2}\overbrace{\cos(w)}^{>0}(\overbrace{u(t)\sin(w)}^{<0}\overbrace{w'}^{>0} - \frac{1}{4}\overbrace{u'(t)}^{\geq 0}\overbrace{\cos(w)}^{>0} - \overbrace{(w')^3}^{>0}) < 0$$

Hence, we know $p''(t) \geq 0$ if $t \in [0, t_p)$, and $p''(t) \leq 0$ if $t \in (t_p, 1]$.

More generally, we know u is of bounded variation because it is increasing, and hence can be considered as a Radon measure. By Lebesgue's Decomposition theorem, we can write $u = u_a + u_s$, where u_a is the absolutely continuous part and has derivative almost everywhere on $[0, 1]$, while u_s is the singular part, capturing the countable discontinuities. Furthermore, in the sense of the distributional derivative [16] , we have: $du = u_a' dt + du_s$, and we know $u_a' \geq 0$ for almost all t and du_s is a non-negative measure on $[0, 1]$.

Meanwhile, our previous regularity result implies that w' is Lipschitz-continuous and hence w'' exists almost everywhere, and we still have:

$$p'' = \frac{1}{2}\cos(w)w'' - \frac{1}{2}\sin w(w')^2$$

Now, for higher-order derivatives, we have:

$$\begin{aligned} dw'' &= \frac{1}{4}u\sin(w)\,w'\,dt - \frac{1}{4}\cos(w)\,du \\ dp'' &= \frac{1}{2}\cos(w)(u\sin(w)w'dt - \frac{1}{4}\cos(w)du(t) - (w')^3 dt) \end{aligned}$$

since du is a non-negative measure, we know dp'' is a non-positive measure restricted on the interval $(\min\{t_u, t_w\}, \max\{t_u, t_w\})$ for the case where w crosses 0; on the interval $[t_u, 1]$ when w stays above 0; on the interval $[0, t_u]$ when w stays below 0. Hence, p'' is weakly decreasing on the interval in each of the above cases.

A.5 Proof of the comparative statics

A recurrent idea behind the proof of Propositions 8–12 is to consider the difference $z = w_2^* - w_1^*$; next,. We show that z satisfies a linear second-order boundary value problem (BVP) by

[16] That is, for any smooth function ϕ supported in $[0, 1]$, i.e. $\phi \in \mathcal{C}_c^\infty([0, 1])$, we have:

$$\int_0^1 \phi(t)du(t) = \int_0^1 \phi(t)u_a'(t)dt + \int_0^1 \phi(t)du_s(t)$$

subtracting the Optimality equations of w_1^* and w_2^*, and then, we verify the sign of z using a comparison principle.

The following identity is used in writing the BVP of z:

$$\begin{aligned} S(\theta) :&= \int_0^1 \sin(w_1^*(\theta) + sz(\theta))ds \\ &= \begin{cases} -\frac{1}{z}(\cos(w_2^*(\theta)) - \cos(w_1^*(\theta))) & \text{if } z(\theta) \neq 0 \\ \sin(w_1^*(\theta)) & \text{if } z(\theta) = 0 \end{cases} \\ \implies \cos(w_2^*(\theta)) - \cos(w_1^*(\theta)) &= -S(\theta)z(\theta) \end{aligned} \tag{2}$$

Moreover, when w_1^*, $w_2^* \in (0, \pi/2)$ on $(c, b]$, the convex combination $(w_1^*(\theta) + sz(\theta)) \in (0, \pi/2)$ as well, so $S(\theta) > 0$ on $(c, b]$.

The following lemma is used to derive the desirable sign of z:

Lemma A.1 (Comparison Principle). *Let $[l, r] \subset \mathbb{R}$ with $l < r$. Let a_2, a_1, a_0 be the coefficients of the BVP. Suppose $a_2, a_0 \in L^1([l, r])$, $a_1 \in L^\infty([l, r])$ and $a_1 \geq 0$ almost everywhere. Let $z \in C^1([l, r])$ with $z' \in AC([l, r])$ satisfy:*

$$-z''(t) - a_2(t)z'(t) + a_1(t)z(t) = a_0(t) \ \ a.e. \ on \ [l, r].$$

Assume the boundary condition is either:

1. *Mixed boundary conditions: $z(l) = 0$, $z'(r) = 0$;*
2. *Neumann boundary conditions: $z'(l) = z'(r) = 0$, and $\int_l^r a_1(s)\, ds > 0$*

Then, if $a_0 \geq 0$ a.e. on $[l, r]$, then $z \geq 0$ on $[l, r]$. Furthermore, if $a_0 > 0$ on a set of positive measure, then:

- *Under the mixed boundary conditions, $z(t) > 0$ for all $t \in (l, r]$ and $z'(l) > 0$;*
- *Under the Neumann boundary conditions: $z(t) > 0$ for all $t \in [l, r]$.*

Proof. Let $\mu(t) = \exp(\int_l^t a_2(s)\, ds) > 0$. The equation becomes $-(\mu z')' + \mu a_1 z = \mu a_0$. Testing it against the negative part $z^- := \max\{-z, 0\}$ and using the boundary conditions gives

$$-\int_l^r \mu |(z^-)'|^2 - \int_l^r \mu a_1 (z^-)^2 = \int_l^r \mu a_0 z^- \geq 0.$$

Thus z^- is constant. The mixed boundary condition, or $\int_l^r a_1 > 0$ in the Neumann case, forces this constant to be zero.

For strictness, let y solve $(\mu y')' = \mu a_1 y$ with $y(l) = 0, y'(l) = 1$ in the mixed case, and $y(l) = 1, y'(l) = 0$ in the Neumann case. Let v solve the same equation with $v(r) = 1, v'(r) = 0$. The integral equations imply $y > 0$ on $(l, r]$ (and on $[l, r]$ in the Neumann case), $v > 0$ on

$[l, r]$, and $y'(r) > 0$. In the Neumann case the last inequality uses $\int_l^r a_1 > 0$. The constant Wronskian $W = \mu(y'v - yv')$ is therefore positive. Variation of constants gives

$$z(t) = \int_l^r \frac{y(\min\{t,s\})v(\max\{t,s\})}{W}\,\mu(s)a_0(s)\,ds.$$

If a_0 is positive on a set of positive measure, positivity of this kernel gives the asserted strict inequalities. In the mixed case, differentiating at l also gives $z'(l) > 0$. □

A.5.1 Proof of Proposition 8 (Stronger incentive sharpens the response)

Proof. Suppose $u_2 \geq u_1 > 0$ on $(c, b]$. Define $z := w_2^* - w_1^*$. By Proposition 5, both w_1^* and w_2^* are interior, strictly increasing and is antisymmetric around c.

Subtracting the Optimality condition for w_1^* from that for w_2^* yields

$$(w_2^* - w_1^*)'' + \frac{g'}{g}(w_2^* - w_1^*)' = -\frac{1}{4\lambda}\big(u_2\cos(w_2^*) - u_1\cos(w_1^*)\big).$$

Adding and subtracting $\frac{1}{4\lambda}u_1\cos(w_2^*)$ and substituting identity (2),

$$(gz')' - \frac{1}{4\lambda}gu_1Sz = -\frac{1}{4\lambda}g\cos(w_2^*)(u_2 - u_1).$$

Dividing by $-g < 0$, we find that z solves the linear BVP on $[c, b]$:

$$-z'' - \frac{g'}{g}z' + \frac{1}{4\lambda}u_1Sz = \frac{1}{4\lambda}\cos(w_2^*)(u_2 - u_1), \quad z(c) = 0, \quad z'(b) = 0,$$

where the boundary conditions follow from symmetry: $w_1^*(c) = w_2^*(c) = 0$ and the boundary condition: $(w_1^*)'(b) = (w_2^*)'(b) = 0$.

On $[c, b]$, $a_1(\theta) := \frac{1}{4\lambda}u_1(\theta)S(\theta) \geq 0$ and $a_0(\theta) = \frac{1}{4\lambda}\cos(w_2^*(\theta))(u_2(\theta) - u_1(\theta)) \geq 0$, since $w_2^* \in (0, \pi/2) \implies \cos(w_2^*) > 0$. By Lemma A.1, $z = w_2^* - w_1^* \geq 0$ on $[c, b]$. If, in addition, $u_2 > u_1$ on a set of positive measure, then $a_0 > 0$ on that set, which by Lemma A.1 implies $z(\theta) > 0$ for all $\theta \in (c, b]$ and $z'(c) > 0$. An identical argument on $[a, c]$ yields $w_2^* \leq w_1^*$, with strict inequality on $[a, c)$ when $u_2 > u_1$ on a positive-measure set.

Because $p(w) = (1 + \sin w)/2$ is strictly increasing on $[-\pi/2, \pi/2]$ and symmetric about $w(c) = 0$, we have $|p_2^*(\theta) - \frac{1}{2}| \geq |p_1^*(\theta) - \frac{1}{2}|$ on $[a, b]$. When $u_2 > u_1$ on a set of positive measure, this inequality is strict for all $\theta \neq c$:

$$\left|p_2^*(\theta) - \frac{1}{2}\right| > \left|p_1^*(\theta) - \frac{1}{2}\right| \quad \text{for all } \theta \neq c.$$

Finally, differentiating $p^*(\theta) = \frac{1}{2}(1 + \sin(w^*(\theta)))$ at $\theta = c$ gives

$$(p^*)'(c) = \frac{1}{2}(w^*)'(c)\cos(w^*(c)) = \frac{1}{2}(w^*)'(c).$$

Since $z'(c) = (w_2^*)'(c) - (w_1^*)'(c) > 0$, we obtain $(p_2^*)'(c) > (p_1^*)'(c)$. □

A.5.2 Proof of Proposition 11 (Prior concentration flattens the response)

Proof. Suppose $\frac{g_2}{g_1}$ is weakly decreasing on $[c,b]$. Denote the difference between the two solutions by $z = w_2^* - w_1^*$. By Proposition 5, both w_1^* and w_2^* are interior, strictly increasing and is antisymmetric around c.

Again, subtracting their corresponding Optimality equations gives:

$$\begin{aligned}
&((w_2^*)'' - (w_1^*)'') + (\frac{g_2'}{g_2}(w_2^*)' - \frac{g_1'}{g_1}(w_1^*)') = -\frac{1}{4\lambda}u\left(\cos(w_2^*) - \cos(w_1^*)\right)\\
\Leftrightarrow &z'' + \frac{g_2'}{g_2}z' + \frac{1}{4\lambda}u(\cos(w_2^*) - \cos(w_1^*)) = (\frac{g_1'}{g_1} - \frac{g_2'}{g_2})(w_1^*)'\\
\Leftrightarrow &z'' + \frac{g_2'}{g_2}z' - \frac{1}{4\lambda}uSz = (\frac{g_1'}{g_1} - \frac{g_2'}{g_2})(w_1^*)'
\end{aligned}$$

Let $\hat{z} = -z$, the symmetric environment implies $\hat{z}$ on $[c,b]$ satisfies:

$$-\hat{z}'' - \frac{g_2'}{g_2}\hat{z}' + \frac{1}{4\lambda}uS\hat{z} = (\frac{g_1'}{g_1} - \frac{g_2'}{g_2})(w_1^*)', \quad \hat{z}(c) = 0, \quad \hat{z}'(b) = 0$$

Taking derivative with the ratio $\frac{g_2}{g_1}$ gives $g_2'g_1 - g_1'g_2 \le 0$ a.e. $\Rightarrow \frac{g_1'}{g_1} - \frac{g_2'}{g_2} \ge 0$ a.e. on $[c,b]$. Besides, u being single-crossing implies $(w_1^*)' > 0$ on $[c,b)$. Applying Lemma A.1 with $\hat{z}$, we know $z = w_2^* - w_1^* \le 0$ on $(c,b]$. If additionally, the likelihood ratio is not constant, $z = w_2^* - w_1^* < 0$ on $(c,b]$ with $z'(c) = (w_2^*)'(c) - (w_1^*)'(c) < 0$.

An identical argument gives the reverse order on $[a,c]$.

Since $p = (1+\sin w)/2$ is increasing in $w \in [-\pi/2, \pi/2]$ and $p'(c) = w'(c)\cos(w(c))/2 = w'(c)/2$, we know $|p_2^*(\theta) - \frac{1}{2}| \le |p_1^*(\theta) - \frac{1}{2}|$. If $\frac{g_2}{g_1}$ is nonconstant, we further have

$$|p_2^*(\theta) - \tfrac{1}{2}| < |p_1^*(\theta) - \tfrac{1}{2}| \text{ when } \theta \neq c, \text{ and } (p_2^*)'(c) < (p_1^*)'(c).$$

□

A.5.3 Proof of Proposition 10

Proof. We will first use a sorting argument to establish the desired weak inequality.[17] Let $p_M = \max\{p_L^*, p_H^*\}$ and $p_m = \min\{p_L^*, p_H^*\}$. The sum of information costs under p_M and p_m equals to the sum under p_L^* and p_H^*:

$$C_{FI}(p_M; g) + C_{FI}(p_m; g) = C_{FI}(p_L^*; g) + C_{FI}(p_H^*; g),$$

because the functions p_M and p_m and their derivative can be matched with p_L^* and p_H^* and their derivatives at almost every state, and the Fisher information cost is an integral determined by the function and its derivative over the state.

Denote the value of the problem under payoff u and choice rule p as $J_u^P[p]$. Since the cost

[17] We need this step because without symmetry, $S(\theta)$ might be negative, we cannot apply the comparison principle directly.

terms cancel out,

$$J^P_{u_L}[p_m] + J^P_{u_H}[p_M] - J^P_{u_H}[p^*_L] - J^P_{u_H}[p^*_H] = \int_\Theta (u_L p_m + u_H p_M - u_L p^*_L - u_H p^*_H)\, g\, d\theta$$
$$= \int_\Theta (u_H - u_L)(p^*_L - p^*_H)_+ \, g\, d\theta \geq 0,$$ [18]

Meanwhile, since the p^*_L and p^*_H are optimal response under u_H and u_L,

$$J^P_{u_l}[p^*_L] = J^P_{u_l}[p_m] \qquad J^P_{u_H}[p^*_H] = J^P_{u_H}[p_M].$$

The uniqueness of the optimal solution implies $p^*_L = p_m$, $p^*_H = p_M$, hence $p^*_H \geq p^*_L$.

For strict inequality, we consider the following case:

Case I: If $p^*_L \equiv 1$ then $p^*_H \equiv 1$ must also be the optimal solution under u_H.

Case II: If $p^*_L \in (0,1)$ and $p^*_H \equiv 1$, then the inequality satisfies trivially.

Case III: If p^*_L and p^*_H are both interior. Define $z := w^*_2 - w^*_1$. Subtracting the Optimality conditions yields:

$$-z'' - \frac{g'}{g}z' + \frac{1}{4\lambda}u_L S z = \frac{1}{4\lambda}\cos(w^*_2)(u_H - u_L), \quad z'(a) = z'(b) = 0.$$

Since $S(\theta) \geq -1$ and $\|u_1\|_\infty < \infty$, we can choose a $H > 0$ large enough so that $H - \frac{1}{4\lambda}u_L S \geq 0$. Adding Hz and subtracting $\frac{1}{4\lambda}u_L S$ on both sides gives:

$$-z'' - \frac{g'}{g}z' + Hz = \frac{1}{4\lambda}\cos(w^*_2)(u_H - u_L) + (H - \frac{1}{4\lambda}u_L S)z, \quad z'(a) = z'(b) = 0.$$

Since $z \geq 0$ and $\cos(w^*_2)(u_H - u_L) > 0$ on some set of positive measure, we can apply Lemma A.1 with $a_0 = \frac{1}{4\lambda}\cos(w^*_2)(u_H - u_L) + (H - \frac{1}{4\lambda}u_L S)z$ to establish that $z(\theta) > 0$ on $[a,b]$. Hence, $p^*_H(\theta) > p^*_L(\theta)$ for every $\theta \in \Theta$. □

A.5.4 Proof of Proposition 12

Proof. **Case I**: If p^*_L is interior, then Optimality condition can be rearranged into:

$$(w^*_L)'' + \frac{g'_L}{g_L}(w^*_L)' = -\frac{1}{4\lambda}u\cos(w^*_L)$$
$$\Rightarrow (w^*_L)'' + \frac{g'_H}{g_H}(w^*_L)' = -\frac{1}{4\lambda}\cos(w^*_L)\left(u - \frac{4\lambda}{\cos(w^*_L)}(\frac{g'_H}{g_H} - \frac{g'_L}{g_L})(w^*_L)'\right)$$

Denote $\tilde{u}(\theta) := u(\theta) - \frac{4\lambda}{\cos(w^*_L(\theta))}(\frac{g'_H(\theta)}{g_H(\theta)} - \frac{g'_L(\theta)}{g_L(\theta)})(w^*_L)'$, which is also bounded and piecewise-continuous. We know w^*_L is also the solution to problem under prior g_H and payoff $\tilde{u}$.

Meanwhile, since $(w^*_L)' > 0$ on (a,b) and $(\frac{g'_H(\theta)}{g_H(\theta)} - \frac{g'_L(\theta)}{g_L(\theta)}) \geq 0$, we know $\tilde{u}(\theta) \leq u(\theta)$, and $p_H \geq p_L$ on $[a,b]$ by Proposition 10. Additionally. if $\frac{g_H}{g_L}$ is nonconstant, $\tilde{u}(\theta) < u(\theta)$ on a set

[18] Here $(p^*_L - p^*_H)_+(\theta) = \begin{cases} p^*_L(\theta) - p^*_H(\theta), & \text{if } p^*_L(\theta) \geq p^*_H(\theta) \\ 0, & \text{otherwise} \end{cases}$.

of positive measure, hence $p_H > p_L$ on $[a, b]$ by Proposition 10.

Case II: If $p_L^* \equiv 0$ and $p_H^* \not\equiv 0$, then the inequality holds trivially.

Case III: If $p_L^* \equiv 1$, we need to show that $p_H^* \equiv 1$ as well. Consider a path of priors defined by:

$$g_s(\theta) = \frac{g_L(\theta)\left(\frac{g_H(\theta)}{g_L(\theta)}\right)^s}{\int_\Theta g_L(\tau)\left(\frac{g_H(\tau)}{g_L(\tau)}\right)^s d\tau}$$

We know that $g_0 = g_L$, $g_1 = g_H$ and g_s is continuous in s in the uniform norm. Moreover, the corresponding optimal response p_s^* is continuous in s in the uniform norm. The argument follows similarly from our proof of existence in Appendix B.5. Uniform convergence of g_s, lower semicontinuity and uniqueness of solution implies that $p_{s_n}^* \to p_s^*$ for $s_n \to s$.

Next, we prove by contradiction that $p_H^* \equiv 1$. Fix a $\theta_0 \in (a, b)$ and $p_0 \in (0, 1)$, and let $\tau := \inf\{s \in [0, 1] : p_s^*(\theta_0) \leq p_0\}$. Because the optimal solution is either a boundary solution or an interior solution according to Theorem 3. We know there exists an interval (α, β) such that $\tau \in (\alpha, \beta)$ and p_s^* is interior for $s \in (\alpha, \beta)$. The argument in **Case I** implies that $p_\tau^* \geq p_s^*$ for all $s \in (\alpha, \beta)$. Meanwhile, uniform continuity of p_s^* in s implies that $p_\alpha^* \equiv 1$. Hence, $p_\tau^* \geq \lim_{s\to\alpha} p_s^* \geq 1$, a contradiction. Therefore $p_H^* \equiv 1$ whenever $p_L^* \equiv 1$. □

B Proof of Results in Section 4

B.1 Technical preliminaries

Notation. Let $J : AC([a,b]) \to \mathbb{R}$ be the integral functional $x \mapsto J[x] := \int_a^b \Lambda\big(\theta, x(\theta), x'(\theta)\big)\, d\theta$. The integrand $\Lambda : \mathbb{R}^3 \to \mathbb{R}$ is called the *Lagrangian*. We use subscripts to denote partial derivatives and superscripts to distinguish between optimization problems (P) and (W). That is:

$$\Lambda_x(\theta,x,v) := \frac{\partial}{\partial x}\Lambda(\theta,x,v) \qquad \Lambda_v(\theta,x,v) := \frac{\partial}{\partial v}\Lambda(\theta,x,v)$$
$$J^P[p] := \int_\Theta \Lambda(\theta,p(\theta),p'(\theta))d\theta \qquad \Lambda^P(\theta,p(\theta),p'(\theta)) := \frac{p'(\theta)^2}{p(\theta)(1-p(\theta))} - u(\theta)p(\theta)$$
$$J^W[w] := \int_\Theta \Lambda(\theta,w,w')d\theta \qquad \Lambda^W(\theta,w(\theta),w'(\theta)) := (w'(\theta))^2 - u(\theta)\frac{1+\sin(w(\theta))}{2}$$

A *solution* refers to an admissible function x^* that satisfies $J[x^*] \geq J[x]$ for all admissible functions x, it is also called as a *minimizer*. We say that a solution is *admissible* if it satisfies the constraints of the problem and the objective value is finite. The admissible classes of function to Problem (P) and (W) are denoted, respectively, as[19]

$$\mathcal{D}^P := \left\{ p \in AC(\Theta,[0,1]) : \int_\Theta \frac{(p'(\theta))^2}{p(\theta)(1-p(\theta))} d\theta < +\infty \right\}$$
$$\mathcal{D}^W := \left\{ w \in AC(\Theta,[-\frac{\pi}{2},\frac{\pi}{2}]) : \int_\Theta (w'(\theta))^2 d\theta < +\infty \right\}$$

Folding lemma. In our reparameterization, we require $w(\theta) \in [-\pi/2, \pi/2]$ to ensure invertibility, which creates a point-wise state constraint for our Problem (W). The following simple lemma shows that the relaxed and constrained problems have the same maximum, so every solution w^* of Problem (W) is also a global maximizer of the relaxed problem.[20] This observation simplifies the proof of Necessity Theorem 1 in Appendix B.4 and Existence Theorem 3 in Appendix B.5.

Lemma B.1. *For $w \in AC(\Theta,\mathbb{R})$, with $\int_\Theta (w')^2 < \infty$, let $\hat{w} := \arcsin(\sin w) \in AC(\Theta,[-\pi/2,\pi/2])$, then $J^W[w] = J^W[\hat{w}]$.*

Proof. Denote the function $T(x) = \arcsin(\sin x)$. It is continuous and differentiable except at

[19] Since u is bounded, the following sets are equivalent:

$$\{p \in AC(\Theta,[0,1]) : \int_\Theta \frac{(p'(\theta))^2}{p(\theta)(1-p(\theta))} d\theta < +\infty\} = \{p \in AC(\Theta,[0,1]) : \int_\Theta \frac{(p'(\theta))^2}{p(\theta)(1-p(\theta))} - u(t)p(\theta)d\theta < +\infty\}$$
$$\{w \in AC(\Theta,[-\frac{\pi}{2},\frac{\pi}{2}]) : \int_\Theta (w'(\theta))^2 d\theta < +\infty\} = \{w \in AC(\Theta,[-\frac{\pi}{2},\frac{\pi}{2}]) : \int_\Theta (w'(\theta))^2 - \frac{u(t)\sin(w(\theta))}{2} d\theta < +\infty\}$$

We drop the second term for notational simplicity.

[20] A relaxed problem refers to the following optimization problem without state constraint:

$$\underset{w \in AC(\Theta,\mathbb{R})}{\text{minimize}} \int_\Theta (w'(\theta))^2 - u(\theta)\frac{1+\sin(w(\theta))}{2}\, d\theta. \tag{3}$$

$x = \frac{\pi}{2} + k\pi, k \in \mathbb{Z}$, with derivative $T'(x) = \text{sgn}(\cos x) = \pm 1$. We know:

$$|(T \circ w)'| = |w'| \quad \text{a.e. on } \Theta$$

Indeed, on $\{\theta \in \Theta : w(\theta) \neq \frac{\pi}{2} + k\pi\}$, this follows from the chain rule; on $\{\theta \in \Theta : w(\theta) = \frac{\pi}{2} + k\pi\}$, $|(T \circ w)'| = |w'| = 0$ almost everywhere except where the fold occurs. Meanwhile, $T(w) = w$, together with $((T \circ w)')^2 = (w')^2$ a.e. on Θ, we know $J^W[w] = J^W[T(w)] = J^W[\hat{w}]$. □

Equivalence of weighted and unweighted L^2 norms.

B.2 Proof of Lemma 4.1 (Bijection)

We argue that it is equivalent to work on problem (W) because there is a bijection between the two admissible classes that preserves the value of the integral. The following is a more formal re-statement of our bijection lemma in the main text:

Lemma B.2 (Restatement of Lemma 4.1). *The transformations $\Phi(p) := \arcsin(2p-1)$ and $\Psi(w) := \frac{1+\sin(w)}{2}$ define a bijection between the two classes of admissible solutions:*

$$p = \frac{1+\sin(w)}{2} \in \mathcal{D}^P \quad \Longleftrightarrow \quad w = \arcsin(2p-1) \in \mathcal{D}^W.$$

Moreover, the value of the objective function is preserved under such a reparametrization, that is

$$\int_\Theta g(\theta)[u(t)p(\theta) - \frac{p'(\theta)^2}{p(\theta)(1-p(\theta))}]\,d\theta = \int_\Theta g(\theta)[u(\theta)\frac{1+\sin(w(\theta))}{2} - (w'(\theta))^2]\,d\theta$$

if $p = \frac{1+\sin(w)}{2}$ and both are admissible.

Proof. ($\Leftarrow$:) If $\boldsymbol{w \in AC(\Theta, [-\frac{\pi}{2}, \frac{\pi}{2}])}$ and $\boldsymbol{\int_\Theta (w'(\theta))^2 dt < \infty}$.

Because $\sin(\cdot)$ is a Lipschitz function and the composition of a Lipschitz function with an absolutely continuous function is again absolutely continuous. We know $p(\theta) = \frac{1+\sin(w(\theta))}{2} \in AC(\Theta)$. Meanwhile, $p(\theta) \cdot (1-p(\theta)) = \frac{1-\sin(w(\theta))^2}{4} = \frac{\cos(w(\theta))^2}{4}$ and $p'(\theta) = \cos(w(\theta)) \cdot w'(\theta)$ is defined almost everywhere on Θ. On $\{\theta : w(\theta) = \pm\pi/2\}$, $p' = w' = 0$ a.e. Hence $\frac{p'(\theta)^2}{p(\theta)(1-p(\theta))} = w'(\theta)^2$ almost everywhere. So, $\int_\Theta \frac{p'(\theta)^2}{p(\theta)(1-p(\theta))} dt = \int_\Theta (w'(\theta))^2 dt < \infty$ and $p \in \mathcal{D}^P$.

($\Rightarrow$:)If $\boldsymbol{p \in AC(\Theta, [0,1])}$ and $\boldsymbol{\int_\Theta \frac{p'(\theta)^2}{p(\theta)(1-p(\theta))} dt < \infty}$.

Consider a sequence $\epsilon_n \to 0$, we truncate p to be $p_{\epsilon_n} = \max\{\min\{p, 1-\epsilon_n\}, \epsilon_n\}$, so $p_{\epsilon_n} \in [\epsilon_n, 1-\epsilon_n]$. Define $w_{\epsilon_n} = \arcsin(2p_{\epsilon_n} - 1)$. Because $\arcsin(\cdot)$ is Lipschitz when bounded away from 0 and 1, $w_{\epsilon_n} \in AC(\Theta)$, with

$$(w'_{\epsilon_n})^2 = \frac{(p')^2}{p(1-p)} \mathbf{1}\{\epsilon_n < p < 1-\epsilon_n\} \text{ a.e.}$$

Hence, $w'_{\epsilon_n} \to \frac{p'}{\sqrt{p(1-p)}} \mathbf{1}\{0 < p < 1\}$ by Dominated convergence theorem in L^2. Meanwhile $p_{\epsilon_n} \to p$ uniformly implies $w'_{\epsilon_n} \to w := \arcsin(2p-1)$ uniformly by uniform continuity of

arcsin(·). Therefore $w(\theta) = w(a) + \int_a^\theta \frac{(p')^2}{p(1-p)} \mathbf{1}\{\epsilon_n < p < 1-\epsilon_n\}\, ds$ is absolutely continuous, with $\int_\Theta (w'(\theta))^2 d\theta = \int_\Theta \frac{p'(\theta)^2}{p(\theta)(1-p(\theta))} d\theta < \infty$, so $w \in \mathcal{D}^W$.

□

B.3 Proof of Lemma 4.2 (Concavity)

Proof. Since the term $u(\theta)p(\theta)$ is linear in p, we only need to prove the convexity of the following function:

$$\phi(x,y) = \begin{cases} \frac{y^2}{x(1-x)} = \frac{y^2}{x} + \frac{y^2}{1-x} & \text{if } x \in (0,1) \\ 0 & \text{if } y = 0 \text{ and } x = 0 \text{ or } 1 \\ +\infty & \text{otherwise .} \end{cases}$$

Notice that its effective domain $\mathrm{dom}\phi := \{(x,y) : \phi(x,y) < +\infty\} = (0,1) \times \mathbb{R} \cup \{0,1\} \times \{0\}$ is a non-empty convex set. We only need to prove the convexity of ϕ on its effective domain.

i) For the case where $(x_1, y_1), (x_2, y_2) \in (0,1) \times \mathbb{R}$, by Cauchy–Schwarz inequality,

$$\frac{(\alpha y_1 + (1-\alpha)y_2)^2}{\alpha x_1 + (1-\alpha)x_2} \le \alpha \frac{y_1^2}{x_1} + (1-\alpha)\frac{y_2^2}{x_2}$$
$$\frac{(\alpha y_1 + (1-\alpha)y_2)^2}{\alpha(1-x_1) + (1-\alpha)(1-x_2)} \le \alpha \frac{y_1^2}{1-x_1} + (1-\alpha)\frac{y_2^2}{1-x_2}$$

where the first equality holds exactly when $\frac{y_1}{x_1} = \frac{y_2}{x_2}$ and second equality holds exactly when $\frac{y_1}{1-x_1} = \frac{y_2}{1-x_2}$, these together require $(x_1, y_1) = (x_2, y_2)$ or $y_1 = y_2 = 0$.

ii) For the case where $(x_2, y_2) = (0,0)$ and $(x_1, y_1) \in (0,1) \times \mathbb{R}$, we have

$$\phi(\alpha x_1 + (1-\alpha)x_2, \alpha y_1 + (1-\alpha)y_2) = \phi(\alpha x_1, \alpha y_1)$$
$$= \frac{\alpha y_1^2}{x_1(1-\alpha x_1)} \le \frac{\alpha y_1^2}{x_1(1-x_1)} = \alpha\phi(x_1,y_1) + (1-\alpha)\phi(x_2,y_2),$$

and the case of $(x_2, y_2) = (1,0)$ and $(x_1, y_1) \in (0,1) \times \mathbb{R}$ is treated symmetrically.

iii) For the case where $(x_1, y_1), (x_2, y_2) \in \{0,1\} \times \{0\}$, we have $\phi(\alpha x_1 + (1-\alpha)x_2, \alpha y_1 + (1-\alpha)y_2) = 0 = \alpha\phi(x_1,y_1) + (1-\alpha)\phi(x_2,y_2)$.

Therefore, at each θ, the convexity inequality holds if and only if $(p_1, p_1') = (p_2, p_2')$ or $p_1' = p_2' = 0$, integrating inequality implies:

$$C_{FI}(\alpha p_1 + (1-\alpha)p_2; g) \le \alpha C_{FI}(p_1 + p_2; g) + (1-\alpha) C_{FI}(p_1 + p_2; g)$$

Finally, since $g > 0$ and $\lambda > 0$, $J^P[\alpha p_1 + (1-\alpha)p_2] = \alpha J^P[p_1] + (1-\alpha)J^P[p_2]$ only if $\phi(\alpha p_1 + (1-\alpha)p_2) = \alpha\phi(p_1, p_1') + (1-\alpha)\phi(p_2')$ almost everywhere, which requires $p_1 = p_2$ or $p_1' = p_2'$ almost everywhere.

□

B.4 Proof of Theorem 1 (Necessity)

Proof. An admissible function w_* is said to be a *strong local minimizer* if it is a minimizer within the neighborhood defined as:

$$J^W[w_*] \leq J^W[w] \text{ for } \forall\, w \in \{w \in AC([0,1],\mathbb{R}) : \|w - w_*\|_\infty \leq \epsilon\}.$$

Certainly, a global minimizer for the unconstrained problem is also a local strong minimizer and satisfies our necessary condition below, which is from Theorem 18.13 of Clarke (2013), and we state it here for completeness.

In the following, $\partial_P \Lambda$ refers to **proximal subdiferential** of $\Lambda(t,x,v)$ with respect to only (x,v),[21] and $\partial_L \Lambda$ refers to **limiting subdifferential** of $\Lambda(t,x,v)$ with respect to only (x,v).[22]

Theorem B.1 (Clarke 2013, Theorem 18.13). *Consider the minimization problem:*

$$J(x) = l(x(a), x(b)) + \int_a^b \Lambda(t, x, x'(t))dt$$

*over $x : [a,b] \to \mathbb{R}^n$, subject to boundary condition $(x(a), x(b)) \in E$, where $E \subset \mathbb{R}^n \times \mathbb{R}^n$ is closed, the Lagrangian Λ is assumed to be finite-valued, **LB** measurable in t and (x,v)*[23] *and l is locally Lipschitz.*

If one further suppose that the Lagrangian Λ satisfies the following condition: for every bounded subset $S \subseteq \mathbb{R}^n$, there exists a constant c and a L_1 integrable function d, such that, for almost all t, for almost all $(x,v) \in S \times \mathbb{R}^n$, one has:

$$\frac{|\zeta|}{1+|\psi|} \leq c(|v| + |\Lambda(t,x,v)|) + d(t), \quad \forall (\zeta,\psi) \in \partial_P \Lambda(t,x,v)$$

*Let x_* be a strong local minimizer of the above problem. There exists an absolutely continuous costate function q satisfying the **Euler inclusion**:*

$$q'(t) \in co\{r : (r, q(t)) \in \partial_L \Lambda(t, x_*(t), x_*'(t))\}\ t \in [a,b] \ \ a.e.,$$

*together with the **Weierstrass condition**: for almost every t,*

$$\Lambda(t, x_*(t), v) - \Lambda(t, x_*(t), x_*'(t)) \geq \langle q(t), v - x_*'(t)\rangle, \ \forall v \in \mathbb{R}^n$$

[21] Let X be a normed space, and $f : X \to \mathbb{R}\bigcup\{+\infty\}$, $x \in \text{dom} f$. We say that $\xi \in X^*$ is a *proximal subgradient* of f at x if for some $\sigma \geq 0$ and for some neighborhood V of x, we have

$$f(y) - f(x) + \sigma(y)\|y - x\|^2 \geq \langle \xi, y - x\rangle \quad \forall y \in V.$$

The *proximal subdifferential* of f at x is the set of proximal subgradient at x.

[22] The *limiting subdifferentiial* is defined as the set of limits of proximal subgradient in the following sense:

$$\partial_L f(x) = \{\xi = \lim \xi_i : \xi_i = \partial_P f(x_i), x_i \to x, f(x_i) \to f(x)\}.$$

[23] In the sense that, it is measurable with respect to the σ-algebra generated by the products of Lebesgue measurable subsets of $[a,b]$ and Borel measurable subsets $\mathbb{R}^2$.

and the ***transversality condition****:*

$$(q(a), -q(b)) \in \partial_L l(x_*(a), x_*(b)) + N_E^L(x_*(a), x_*(b)).$$

First, we verify that the hypothesis of the theorem is satisfied. In our context, there is no endpoint constraint for the value of w, so we can set E to be $\mathbb{R} \times \mathbb{R}$ and $l(w(a), w(b)) = 0$ for any $(w(a), w(b))$. Besides, our Lagrangian $\Lambda^W(t, w, w') = (w')^2 - u(t)\frac{1+\sin(w)}{2}$ is finite on $[a, b]$ for each $w \in AC([0, 1], [-\frac{\pi}{2}, \frac{\pi}{2}])$. It is a **Carathéodory function**, as Λ^W is Lebesgue measurable in t for any absolutely continuous w and it is continuous in $(w(\theta), w'(\theta))$ for each fixed t. Hence, we know it is **LB** measurable by Clarke 2013, Proposition 6.35. Furthermore, for each fixed t, we notice that $\Lambda^W(t, w, w')$ is smooth in (w, w'), so whenever it is non-empty, its **proximal subdifferential** ∂_P coincides with the its gradient (Clarke, 2013, Corollary 7.32), given by

$$(\Lambda_w^W(t, w, w'),\ \Lambda_{w'}^W(t, w, w')) = (\frac{1}{2}u(t)\cos(w),\ 2w'),$$

We can choose $c = 0$, $d(t) = |u(t)|$, which is L_1 integrable and satisfy the inequality:

$$\frac{|\zeta|}{1 + |\psi|} = \frac{\frac{1}{2}|u(t)\cos(w)|}{1 + |2w'|} \leq |u(t)| = c(|w'| + |\Lambda^W(t, w, w')|) + d(t)$$

for $\forall(\zeta, \psi) \in \partial_P \Lambda(t, x, v)$ and all $t \in [a, b]$ and all $(w, w') \in \mathbb{R}^2$.

Next, we interpret each of the necessary conditions.

Since $\Lambda^W(t, w, w')$ is continuously differentiable in (w, w'), the **limiting subdifferential** $\partial_L \Lambda^W$ reduces to its gradient, which is a singleton (Clarke, 2013, Proposition 11.12). So our **Euler inclusion** condition actually implies the **Euler equation**:

$$\begin{aligned} q'(t) = \Lambda_w^W(t, w, w') = -\frac{1}{2}u(t)\cos(w(\theta)), \quad q(t) = \Lambda_{w'}^W(t, w, w') = 2w'(\theta) \\ \Rightarrow w'(\theta) = c + \int_0^t \frac{-1}{4}u(\tau)\cos(w(\tau))d\tau \quad \text{for some constant } c \end{aligned}$$

Besides, the **Weierstrass condition** is automatically satisfied, as Λ^W is convex in w':

$$\Lambda^W(t, w_*(t), v) - \Lambda^W(t, w_*(t), w_*'(t)) \geq \langle \Lambda_{w'}^W(t, w_*(t), w_*'(t)), v - w_*'(t) \rangle,\ \forall v \in \mathbb{R}$$

Lastly, since there is no endpoint constraint for w, we let $l(w(a), w(b)) \equiv 0$ and $E = \mathbb{R} \times \mathbb{R}$, so the limiting subdifferential is

$$\partial_L l(w_*(a), w_*(b)) = \{(0, 0)\},$$

and the limiting normal cone is also a singleton:

$$N_E^L(w_*(a), w_*(b)) = \{(0, 0)\},$$

Our **Transversality condition** implies:

$$(q(0), -q(1)) = (2w'(0), -2w'(1)) = (0, 0).$$

Together with the fact that the above costate function q is absolutely continuous, we have:

$$\begin{cases} w'(\theta) = -\frac{1}{4}\int_0^t u(s)\cos(w(s))ds & \text{a.e. on } [0,1] \\ w'(t_i^-) = w'(t_i^+) & \text{for } = 1, 2, \cdots N-1 \\ w'(0) = w'(1) = 0 \end{cases} \tag{4}$$

□

B.5 Proof of Theorem 3 (Existence)

B.5.1 Existence of AC solution

We will apply the direct method in calculus of variations to establish the existence of an absolutely continuous solution. The method shares the same spirit of Weierstrass' extreme value theorem, which says that a lower semicontinuous function attains its infimum on a compact support. Compactness comes from the coercive condition and proper choice of the topology, while lower semi-continuity is established by the **Integral Semicontinuity** lemma from Clarke (2013, Theorem 6.38), which we state without proof.

Let Ω be an open subset of $\mathbb{R}^m$, and $F : \Omega \times \mathbb{R}^l \times \mathbb{R}^n \to \mathbb{R}$ is the integrand, whose arguments are denoted by (t, x, v). We also use $Q \subseteq \Omega \times \mathbb{R}^l$ to describe the constraint on function $x : \Omega \to \mathbb{R}^l$, i.e. the state constraint:

$$(t, x(t)) \in Q \subseteq \Omega \times \mathbb{R}^l \;\; t \in \Omega \; a.e.$$

We want to show the semicontinuity of the integral functional

$$J(x, v) = \int_\Omega F(t, x(t), v(t))dt.$$

Lemma B.3 (Integral Semicontinuity). *Suppose the following hypotheses holds:*

(a) F is lower semicontinous in (x, v) and convex with respect to v;

(b) For every measurable $x : \Omega \to \mathbb{R}^l$ having $(t, x(t)) \in Q$ a.e., and for every measurable $v : \Omega \to \mathbb{R}^n$, the function $t \mapsto F(t, x(t), v(t))$ is measurable;

(c) There exists $\alpha \in L^1(\Omega)$, $\beta \in L^\infty(\Omega, \mathbb{R}^n)$ such that

$$F(t, x, v) \geq \alpha(t) + \langle \beta(t), v \rangle \;\; \forall (t, x) \in Q, v \in \mathbb{R}^n$$

(d) Q is closed in $\Omega \times \mathbb{R}^l$.

Let $\{x_i\}$ be a sequence of measurable functions on Ω satisfying $(t, x(t)) \in Q$ a.e. which converges almost everywhere to a limit x_*. Let $\{v_i\}$ be a sequence of functions converging weakly in $L^r(\Omega, \mathbb{R}^n)$ to v_*, where $r > 1$. Then

$$J(x_*, v_*) \leq \liminf_{i\to\infty} J(x_i, v_i)$$

Proof of the existence of AC solution. Firstly, we notice that the infimum of the value of problem (W) is finite. This is because our Lagrangian is bounded from below:

$$\Lambda^W(t, x, v) = v^2 - u\frac{1+\sin(x)}{2} \geq -\|u\|_\infty > -\infty.$$

Meanwhile, the constant 0 function achieves a finite $J^W[\mathbf{0}] = 0$. Therefore, there exists a sequence of absolutely continuous function w_i such that

$$\lim_{i\to\infty} J^W[w] = \inf_{w\in AC([0,1],[-\frac{\pi}{2},\frac{\pi}{2}])} J^W[w]$$

Besides, our Lagrangian satisfies the following coercive condition:

$$\Lambda^W(t, x, v) = (v)^2 - u(t)\frac{1+\sin(x)}{2} \geq |v|^2 - \|u\|_\infty \ \ \forall (t, x, v) \in [0, 1] \times [-\frac{\pi}{2}, \frac{\pi}{2}] \times \mathbb{R},$$

For i large enough, we have:

$$\int_0^1 (|w_i'(t)|^2 - \|u\|_\infty)\, dt \leq \int_0^1 \Lambda^W(t, w_i(t), w_i'(t))\, dt \leq \inf_{w\in AC([0,1],[-\frac{\pi}{2},\frac{\pi}{2}])} J^W[w] + 1$$

This implies that the sequence of derivatives $\{w_i'\}$ is a bounded sequence in $L^2[0, 1]$. Every bounded sequence in $L^2[0, 1]$ has a subsequence that converges weakly in $L^2[0, 1]$. To see this, note that $L^2[a, b]$ is a reflexive space, and Banach–Alaoglu theorem says that the closed unit ball in the dual space of a normed vector space is compact in the weak-* topology[24]. Hence, every closed unit ball in $L^2[0, 1]$ is also compact in the weak-* topology on $L^2[0, 1]$. Then, by Eberlein-Šmulian Theorem, we know that in a normed space, weak compactness coincides with weak sequential compactness, so a subset K being compact in the weak topology implies that every sequence in K has a subsequence that converges weakly to some point in K. Lastly, a bounded sequence is contained by some closed ball in $L^2[0, 1]$ and must admit a convergent subsequence in $L^2[0, 1]$. Without loss of generality we can replace the sequence with that subsequence and denote the limit as v_*. Besides, we have the unilateral state constraint:

$$w(\theta) \in [-\frac{\pi}{2}, \frac{\pi}{2}],\ t \in [0, 1].$$

[24]Let $(X, Y, \langle, \rangle)$ be a pairing of topological vector space, with the duality $(x, y) \mapsto \langle x, y\rangle$ from $X \times Y$ to $\mathbb{R}$. The **weak topology** on X induced by Y (denoted $\sigma(X, Y)$) is the coarsest topology on X that makes the mappings $\langle \cdot, y\rangle : X \to \mathbb{R}$ continuous for all $y \in Y$. Equivalently, it is characterized by the **weak convergence** $\overset{w}{\to}$: for any net, $x_\alpha \to x \iff \langle x_\alpha, y\rangle \to \langle x, y\rangle,\ \ \forall y \in Y$; When $Y = X^*$ is the continous dual of X, the **weak-* topology** on Y is the coarsest topology on Y making the mappings $\langle x, \cdot\rangle : Y \to \mathbb{R}$ continuous for all $x \in X$. Equivalently, it is characterized by the **weak-* convergence** $\overset{w^*}{\to}$: for any net, $y_\alpha \to y \iff \langle x, y_\alpha\rangle \to \langle x, y\rangle,\ \ \forall x \in X$.

Since $[-\frac{\pi}{2}, \frac{\pi}{2}]$ is a compact subset of $\mathbb{R}$, it is also without loss of generality to assume $\lim_{i\to\infty} w_i(0)$ exists.

Together, we define an absolutely continuous w_* as following:

$$w_*(t) = \lim_{i\to\infty} w_i(0) + \int_0^t v_*(s)ds, \quad t \in [0,1],$$

It can be shown that the sequence $\{w_i\}$ converge pointwise to w_*, by choosing the characteristic function as test function. Let $\mathbf{1}_{(a,t)}(s) := \begin{cases} 1 & s \in (a,t) \\ 0 & \text{otherwise} \end{cases}$, by weak-* convergence:

$$\begin{aligned} &\lim_{i\to\infty} \int_0^1 w_i'(s)\mathbf{1}_{(0,t)}(s)ds = \int_0^1 v_*(s)\mathbf{1}_{(a,t)}(s)ds \\ \Rightarrow &\lim_{i\to\infty} w_i(t) = \lim_{i\to\infty} w_i(0) + \int_0^t w_i'(s)ds = w_*(0) + \int_a^t v_*(s)ds = w_*(t) \end{aligned}$$

Since each w_i satisfies the state constraint, and S is closed, we know the state constraint is satisfied by w_* at each point:

$$w_*(t) = \lim_{i\to\infty} w_i(t) \in S, \quad t \in [0,1].$$

Then, we will verify that each hypothesis of the **Integral Semicontinuity** lemma is satisfied. For (a), it is easy to see our Lagrangian $\Lambda^W(t,x,v)$ is continuous in (x,v) and convex in v. For (b), u being piecewise continuous implies $\Lambda^W(t,x,v)$ is measurable in t, together with $\Lambda^W(t,x,v)$ being continuous in (x,v), we know the mapping $t \mapsto \Lambda^W(t, w_i(t), w_i'(t))$ is measurable by Clarke (2013, Prop. 6.34, 6.35). For (c), we can choose $\alpha(t) = -\|u\|_\infty$ and $\beta(t) = 0$, they satisfy $\alpha \in L^1([0,1])$, $\beta(x) \in L^\infty([0,1])$ and $\Lambda^W(t,x,v) \geq -\|u\|_\infty \ \forall (t,u) \in Q, v \in \mathbb{R}$. For (d), we take $Q = [0,1] \times [-\frac{\pi}{2}, \frac{\pi}{2}]$, and it is closed in $\Omega \times \mathbb{R}$.

Lastly, $\{w_i\}$ converges pointwise to w_* implies it converges almost everywhere to w_*, and $\{w_i'\}$ also converges weakly to v_* in $L^2[0,1]$, by Lemma B.3, we know that the infimum is achieved by w_*:

$$J^W[w_*] \leq \lim_{i\to\infty} J^W[w_i] = \inf_{w \in AC([0,1],[-\frac{\pi}{2},\frac{\pi}{2}])} J^W[w].$$

Since w_* is an absolutely continuous function by construction and satisfies the unilateral state constraint, it is indeed a global minimizer for problem (W). □

B.5.2 Existence of Lipschitz solution

Next, we establish the existence of a solution in the class of Lipschitz continuous functions through regularity induced by the Euler integral equation:

Corollary 4. *Problem (W) has a solution in Lip*$([0,1], [-\frac{\pi}{2}, \frac{\pi}{2}])$.

Proof. Since u is a piecewise continuous function on a compact support, we know it is bounded by some constant M almost everywhere except at finite points. Meanwhile, there exists an

absolutely continuous solution that satisfies the Euler Integral Equation:

$$w'(\theta) = \int_0^t \frac{1}{4} u(s)\cos(w(s))ds \qquad \text{a.e. on } [0,1]$$
$$\Rightarrow |w'(s) - w'(\theta)| = |\int_t^s u(\tau)\cos(w(\tau))\,d\tau| \leq M|t-s|,$$

so we know w' is uniformly bounded on [0,1] and hence $w \in Lip$. □

B.5.3 Non-existence of partial boundary solution

The idea of the proof is to first consider the problem without state constraint, where a global minimizer exists, and the extremals satisfy the set of necessary conditions described in Theorem 1. We then argue that any candidate that stays at the boundary and then leaves cannot be an extremal in the unconstrained problem. Later, we show that any candidate that crosses the boundary cannot be a global minimum of the unconstrained problem. Hence, it is a contradiction if a partial boundary candidate can be the global minimizer in the constrained problem.

(Existence of solution to the unconstrained problem): More specifically, we first consider the following optimization problem without state constraint:

$$\underset{w\in AC(\Theta,\mathbb{R})}{\text{minimize}} \int_\Theta (w'(\theta))^2 - u(\theta)\frac{1+\sin(w(\theta))}{2}\,d\theta. \qquad \text{(relaxed-}W\text{)}$$

We first observe the functional is translation-invariant in the sense that $J^W[w] = J^W[w + 2\pi]$. As a result, we may lose compactness due to translation. To account for this, we consider two admissible functions w_1, w_2 as equivalent if $w_1(t) - w_2(t) = 2k\pi$ *a.e.* for some $k \in \mathbb{Z}$, and pick a representative in each equivalent class that satisfies $w(a) \in [-\pi, \pi)$. The existence proof for an $AC(\Theta, \mathbb{R})$ optimal solution proceeds almost exactly as in the case with state constraint, with the only twist that $[-\pi, \pi)$ is closed in the quotient space $\mathbb{R}/2\pi\mathbb{Z}$ and hence $\lim_{i\to\infty} w_i(a)$ exists and we can construct w_* from the minimizing sequence in the exact same way.

The solution, as a global minimizer, satisfies the necessary conditions in Theorem 1 and is governed by the following boundary value problem (BVP)

$$\begin{cases} w''(t) = -\frac{1}{4}u(t)\cos(w_*(t)) & \text{a.e. on } [0,1] \\ w'(0) = w'(1) = 0 \end{cases} \tag{5}$$

(Implication of the necessary conditions): The above equations do not admit solutions that stay at the boundary $\pm\frac{\pi}{2}$ and then leave. To see this, we assume that w is a solution to the BVP and $w(\theta) = -\frac{\pi}{2}$ on some interval $(t_0, t_1) \subsetneq [0,1]$. Then, by the continuity of w and

w', we know $w(t_1) = -\frac{\pi}{2}$ and $w'(t_1) = 0$. We get the following initial value problem (IVP)

$$\begin{aligned} &\begin{cases} y_1(t) = w(\theta) \\ y_2(t) = w'(\theta) \end{cases} \qquad \begin{cases} y_1'(t) = y_2(t) \\ y_2'(t) = -\frac{1}{4}u(t)\cos(y_1(t)) \end{cases} \\ &(y_1(t_1), y_2(t_1)) = (-\frac{\pi}{2}, 0) \\ \Rightarrow &y' = F(t,y), \quad F(t,y) := (y_2(t), -\frac{1}{4}u(t)\cos(y_1(t))) \end{aligned} \tag{6}$$

We note that $F : [0,1] \times \mathbb{R}^2 \to \mathbb{R}^2$ is a **Carathéodory function**: it is Lebesgue measurable in t for each y and continuous in (y_1, y_2) for each t. Moreover, F is Lipschitz in y with a Lipschitz constant independent of t in the strip $[0,1] \times [-\pi, \pi] \times \mathbb{R}$:

$$\begin{aligned} \|F(t,y) - F(t,\tilde{y})\| &= \|(\tilde{y}_2 - y_2, -\frac{1}{4}u(t)(\cos(y_1) - \cos(\tilde{y}_1)))\| \\ &= \|(\tilde{y}_2 - y_2, -\frac{1}{4}\sin(c)u(t)(y_1 - \tilde{y}_1))\| \text{ for some } c \in (y_1, \tilde{y}_1) \\ &\leq \max_{t\in[0,1]} |u(t)| \, \|y - \tilde{y}\| \end{aligned}$$

By a version of **Picard-Lindelöf** Theorem (Walter, 1998, §10.XVIII), we know there exists a unique solution to the IVP. Meanwhile, it is easy to see that $w \equiv -\pi/2$ and $w' \equiv 0$ on $[t_1, 1]$ solve the IVP. The argument for $[0, t_0]$ and for $w = \frac{\pi}{2}$ is almost identical and we omit it here. Hence, if $w(\theta) = -\frac{\pi}{2}$ on some interval (t_0, t_1), it must be on the boundary for the whole interval $[0,1]$. Therefore, any extremal of the unconstrained problem must be one of the following three types up to addition or subtraction of multiple of 2π:

- Pure boundary extremal: $w(\theta) \equiv -\frac{\pi}{2}$ on $[0,1]$ or $w(\theta) \equiv \frac{\pi}{2}$ on $[0,1]$
- Strict interior extremal: $-\frac{\pi}{2} < w(\theta) < \frac{\pi}{2}$ for all $t \in [0,1]$
- Boundary-crossing extremal: $w(\theta) \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ on a subset of $[0,1]$ of positive measure, and $w(\theta) \notin \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ on another subset of positive measure, while the contact set $\{t \in [0,1] : |w(\theta)| = \frac{\pi}{2}\}$ has measure zero.

(Nonexistence of boundary-crossing minimizer): We proceed to argue that, any boundary-crossing extremal, if it ever exists, cannot be a global minimizer of the unconstrained problem. Since J^W is invariant under addition of $2k\pi$, the unconstrained problem is naturally posed on $\mathbb{R}/(2\pi\mathbb{Z})$. For the compactness argument we identify functions that differ by a constant multiple of 2π, and select a representative with $w(0) \in [-\pi, \pi)$. In the arguments below, we work with this fixed representative, so all interval comparisons are made in $\mathbb{R}$, not on the quotient. Recall Assumption **??** of u being single crossing, we have the following useful observations:

Observation B.1. *If for some integer k, $w \in [-\pi + 2k\pi, -\frac{\pi}{2} + 2k\pi)$ on some non-trivial interval on $[0, t_u)$ or $w \in (\frac{\pi}{2} + 2k\pi, \pi + 2k\pi]$ on some non-trivial interval on $(t_u, 1]$, then w cannot be a global minimizer.*

Proof. Suppose $k = 0$ and $w \in [-\pi, -\frac{\pi}{2})$ on some maximal interval $(t_0, t_1) \subseteq [0, t_u)$ with two possible cases: i) $w(t_0) = w(t_1) = -\frac{\pi}{2}$ and $t_0 > 0$, or ii) $w(t_1) = -\frac{\pi}{2} > w(t_0)$ and $t_0 = 0$, such an interval exists because w is continuous, we then construct

$$\tilde{w}(t) = \begin{cases} -\frac{\pi}{2} & \text{on } (t_0, t_1) \\ w & \text{otherwise} \end{cases}$$

the new function $\tilde{w}$ is still absolutely continuous and strictly dominates w:

$$J^W[w] - J^W[\tilde{w}] = \int_{t_0}^{t_1} (w')^2 + \overbrace{\frac{u}{4}}^{<0} \overbrace{(\sin(-\frac{\pi}{2}) - \sin(w))}^{<0} \, dt > 0$$

Therefore, w cannot be a global minimizer. The proof for $w \in (\frac{\pi}{2}, \pi]$ on some non-trivial interval on $(t_u, 1]$ is almost identical and the extension to arbitrary k follows from 2π-translation invariance of J^W □

Observation B.2. *If up to addition or subtraction of some multiple of 2π, $w(t_u) \notin [-\frac{\pi}{2}, \frac{\pi}{2}]$, then w cannot be a global minimizer.*

Proof. Suppose $w(t_u) \in [-\pi, -\frac{\pi}{2})$, then by the continuity of w, there exist an open interval in $[0, t_u)$ such that $w(\theta) \in [-\pi, -\frac{\pi}{2})$ on that interval, and according to our above observation, it cannot be a global minimizer. The case for $w(t_u) \in (\frac{\pi}{2}, \pi]$ is almost identical. □

With these two observation, we only need to consider extremals that satisfy $w(\theta) \in [-\frac{\pi}{2}, \pi]$ on $[0, t_u)$, $w(\theta) \in [-\pi, \frac{\pi}{2}]$ on $(t_u, 1]$ and $w(t_u) \in [-\frac{\pi}{2}, \frac{\pi}{2}]$. Now, suppose that $w \in (\frac{\pi}{2}, \pi]$ on some interval in $[0, t_u)$, we will first consider the following reflection:

$$\tilde{w}(t) = \begin{cases} \pi - w & \text{if } w \in (\frac{\pi}{2}, \pi] \ (t_0, t_1) \\ w & \text{otherwise} \end{cases}$$

We know $(w')^2 = (\tilde{w}')^2$ almost everywhere, and combining with the symmetry of $\sin(\cdot)$ function around $\frac{\pi}{2}$, we also have $\sin(w) = \sin(\tilde{w})$ for all t, this implies $J^W[w] = J^W[\tilde{w}]$. We will show that $\tilde{w}$ is strictly dominated by the following construction:

$$\hat{w} = \begin{cases} \min\{\tilde{w}, \frac{\tilde{w}(t_u) - \tilde{w}(0)}{t_u} t + \tilde{w}(0)\} & \text{on } [0, t_u) \\ w & \text{on } [t_u, 1] \end{cases}$$

Since affine function $w = A + \frac{B-A}{t_1 - t_0} t$ is the solution to the following problem:

$$\begin{aligned} &\text{minimize} \int_{t_0}^{t_1} (w'(\theta))^2 dt \\ &\text{subject to } \ w(t_0) = A, \quad w(t_1) = B, \end{aligned}$$

we know $\int_{t_2}^{t_3} (\tilde{w}'(t))^2 - (\hat{w}'(t))^2 dt > 0$ within each interval where the two disagree, that is

$(t_2, t_3) \in (t_0, t_1)$ where we have $\tilde{w}(t_0) = \hat{w}(t_0)$ and $\tilde{w}(t_1) = \hat{w}(t_1)$ and $\tilde{w} \neq \hat{w}$ on (t_0, t_1). Meanwhile, we have $-\frac{\pi}{2} \leq \hat{w} \leq \tilde{w} \leq \frac{\pi}{2}$, which implies $\sin(\hat{w}) \leq \sin(\tilde{w})$ on $[0, t_u]$. We therefore have

$$\begin{aligned} &J^W[w] - J^W[\hat{w}] = J^W[\tilde{w}] - J^W[\hat{w}] \\ &= \int_0^{t_u} \overbrace{(\tilde{w}'(t))^2 - (\hat{w}'(t))^2}^{>0} - \overbrace{\frac{u(t)}{4}}^{<0} \overbrace{(\sin(\tilde{w})(t) - \sin(\hat{w})(t))}^{\geq 0} dt > 0 \end{aligned}$$

The case for $w \in [-\pi, \frac{\pi}{2})$ on some interval in $(t_u, 1]$ is similar, we first reflect the part $[-\pi, \frac{\pi}{2})$ around $-\frac{\pi}{2}$ to construct a $\tilde{w}$, then let

$$\hat{w} = \begin{cases} \max\{\tilde{w}, \frac{\tilde{w}(t_u) - \tilde{w}(1)}{t_u - 1} t + \tilde{w}(1)\} & \text{on } (t_u, 1] \\ w & \text{on } [0, t_u] \end{cases}$$

Therefore, we have shown that any boundary-crossing extremal cannot be a global minimizer. **(Nonexistence of partial boundary minimizer):** Now, suppose that for the constrained problem, the global minimizer is some w^* such that $w^* = \frac{\pi}{2}$ on some strict subset of $[0, 1]$. Together with our uniqueness result (Corollary 4.2), this implies that neither pure boundary extremal or strict interior extremal can be a global minimizer for the unconstrained problem. Hence, by the existence of the global minimizer to the unconstrained problem, we know it must be a boundary-crossing extremal, contradicting to what we have just shown above.

B.6 Proof of Theorem 4 (Interiority)

Proof. Step 1: a global inequality. Let w be admissible, and put $z := w + \pi/2 \in [0, \pi]$ and $f := \sin(z/2) \in [0, 1]$. Then f is absolutely continuous with $f' = \frac{1}{2}\cos(z/2)\, w'$ a.e., so $(f')^2 \leq (w')^2/4$ and $f \in H^1([a, b])$; moreover $\frac{1}{2}(1 + \sin w) = \frac{1}{2}(1 - \cos z) = f^2$. Since $J[w^L] = 0$,

$$J[w] - J[w^L] = \int_\Theta g\big[uf^2 - \lambda(w')^2\big]d\theta \;\leq\; \int_\Theta g\big[uf^2 - 4\lambda(f')^2\big]d\theta = Q_L[f]. \tag{7}$$

Symmetrically, with $f := \sin\big((\pi/2 - w)/2\big)$ one has $\frac{1}{2}(1 + \sin w) = 1 - f^2$ and $J[w] - J[w^R] \leq Q_R[f]$.

Step 2: sufficiency. Suppose $m_L \leq 0$. For admissible $w \neq w^L$ the function f in Step 1 is not identically zero, so by homogeneity $Q_L[f] \leq m_L \int_\Theta g f^2 \leq 0$, and (7) gives $J[w] \leq J[w^L]$. Hence w^L is optimal. In the same way $m_R \leq 0$ implies that w^R is optimal.

Step 3: necessity. Suppose $m_L > 0$ and pick h with $\int_\Theta g h^2 = 1$ and $Q_L[h] > 0$; replacing h by $|h|$ we may take $h \geq 0$. Since $|\sin^2 x - x^2| \leq x^4/3$ and $\int_\Theta g h^4 \leq \|h\|_\infty^2$, the remainder in (**??**) is at most $\delta^2 \|u\|_\infty \|h\|_\infty^2/3$, so $J[w^L_{\delta,h}] > J[w^L]$ for all small $\delta > 0$ and w^L is not optimal. Likewise $m_R > 0$ implies that w^R is not optimal. Together with Step 2 this shows that w^L is optimal if and only if $m_L \leq 0$, and w^R is optimal if and only if $m_R \leq 0$.

Step 4: the trichotomy. By Corollary 3 the optimal rule is unique, so (i) and (ii) cannot both hold. If neither holds, neither pure rule is optimal; by Theorem 3 an optimal rule exists and is either pure or strictly interior, hence strictly interior. Theorem 1 and Corollary 3 then give the remaining assertions of (iii). □

B.7 Proof of Theorem 2 (Regularity)

Proof. A constant boundary solution $p_* = 0$ or $p_* = 1$ is smooth, and the conclusion is trivially satisfied. Now, suppose the solution is not a boundary one, let $w_* = \arcsin(2p_* - 1)$ be the solution to Problem (W), it has to satisfy the integral Euler equation in (NC). Suppose u is piecewise continuous, then it is also bounded, that is $|u(t)| < M$ for some constant M and for all t. Therefore

$$|(w^*)'(s) - (w^*)'(t)| = |-\frac{1}{4}\int_t^s u(\tau)\cos(w_*(\tau))\,d\tau| \leq M|t-s|,$$

so $(w^*)'$ is Lipschitz continuous and hence it is differentiable almost everywhere, implying that w_* is twice differentiable almost everywhere.

If we further suppose u is continuous, the fundamental theorem of calculus and continuity of $u(t)\cos(w(\theta))$ then also implies:

$$(w^*)''(t) = \lim_{h\to 0} \frac{-\frac{1}{4}\int_t^{t+h} u(\tau)\cos(w_*(\tau))\,d\tau}{h} = -\frac{1}{4}u(t)\cos(w_*(t)).$$

Higher order regularity of w_* can be similarly derived from the the identity $(w^*)''(t) = -\frac{1}{4}u(t)\cos(w_*(t))$. Lastly, since the function $\sin(\cdot)$ is smooth, the result hold equally for p_*. □

C Other proofs

C.1 Fisher information cost is Blackwell monotone

Consider two signal structures $\sigma : \Theta \to S$ and $\hat{\sigma} : \Theta \to \hat{S}$, both with finite signal space and absolutely continuous mappings $\theta \mapsto \sigma(s|\theta)$ and $\theta \mapsto \hat{\sigma}(s|\theta)$ for each s. Suppose σ is more informative than $\hat{\sigma}$ in the sense of Blackwell (1951), that is, there exists a state-independent garbling $K : S \to \Delta(\hat{S})$, such that $\hat{\sigma}(\hat{s}|\theta) = \sum_{s\in S} K(\hat{s}|s)\sigma(s|\theta)$. We will show that experiment $\hat{S}$ is cheaper by showing that the Fisher information is lower almost everywhere, that is $\mathcal{I}_F(\{\hat{\sigma}_\theta\}) \leq \mathcal{I}_F(\{\sigma_\theta\})$ for almost all $\theta \in \Theta$.

To see this, recall that the Fisher information is the variance of the *score*, and the score is the partial derivative with respect to θ of the log-likelihood: $\frac{\partial}{\partial\theta}\log\sigma(s|\theta)$. This is because expectation of the score is 0: since for each $s \in S$, $\theta \mapsto \sigma(s|\theta)$ is absolutely continuous, so for

almost all θ, $\sigma(s|\theta)=0 \Rightarrow \frac{\partial}{\partial\theta}\sigma(s|\theta)=0$, and

$$E_{s|\theta}[\frac{\partial}{\partial\theta}\log\sigma(s|\theta)] = \sum_{s\in S:\sigma(s|\theta)>0}(\frac{\partial}{\partial\theta}\log\sigma(s|\theta))\sigma(s|\theta) = \sum_{s\in S:\sigma(s|\theta)>0}\frac{\frac{\partial}{\partial\theta}\sigma(s|\theta)}{\sigma(s|\theta)}\sigma(s|\theta)$$
$$= \sum_{s\in S}\frac{\frac{\partial}{\partial\theta}\sigma(s|\theta)}{\sigma(s|\theta)}\sigma(s|\theta) = \frac{\partial}{\partial\theta}(\sum_{s\in S}\sigma(s|\theta)) = 0$$

Meanwhile, under the garbling, the score for $\hat{\sigma}(\hat{s}|\theta)$ is the conditional expectation of the score for $\sigma(s|\theta)$: for $\hat{s}\in\hat{S}$ and θ such that $\hat{\sigma}(\hat{s}|\theta)>0$,

$$\frac{\partial}{\partial\theta}\log\hat{\sigma}(\hat{s}|\theta) = \frac{\frac{\partial}{\partial\theta}\hat{\sigma}(\hat{s}|\theta)}{\hat{\sigma}(\hat{s}|\theta)} = \frac{\partial}{\partial\theta}\left(\sum_{s\in S:\sigma(s|\theta)>0}K(\hat{s}|s)\sigma(s|\theta)\right)\frac{1}{\hat{\sigma}(\hat{s}|\theta)}$$
$$= \sum_{s\in S:\sigma(s|\theta)>0}\left(\frac{\partial}{\partial\theta}\sigma(s|\theta)\right)\frac{K(\hat{s}|s)}{\hat{\sigma}(\hat{s}|\theta)} = \sum_{s\in S:\sigma(s|\theta)>0}\frac{\frac{\partial}{\partial\theta}\sigma(s|\theta)}{\sigma(s|\theta)}\frac{K(\hat{s}|s)\,\sigma(s|\theta)}{\hat{\sigma}(\hat{s}|\theta)}$$
$$= \sum_{s\in S:\sigma(s|\theta)>0}\frac{\partial}{\partial\theta}\log\sigma(s|\theta)Pr[s|\hat{S}=\hat{s},\theta] = E_{s|\theta}[\frac{\partial}{\partial\theta}\log\sigma(s|\theta)|\hat{S}=\hat{s}]$$

By conditional Jensen's inequality, the Fisher information for $\hat{\sigma}(\theta)$ is:

$$\mathcal{I}_F(\hat{\sigma}(\theta)) = E_{\hat{s}|\theta}\left[\left(\frac{\partial}{\partial\theta}\log\hat{\sigma}(\hat{s}|\theta)\right)^2\right] = E_{\hat{s}|\theta}\left[\left(E_{s|\theta}[\frac{\partial}{\partial\theta}\log\sigma(s|\theta)|\hat{S}=\hat{s}]\right)^2\right]$$
$$\leq E_{\hat{s}|\theta}\left[E_{s|\theta}\left[\left(\frac{\partial}{\partial\theta}\log\sigma(s|\theta)\right)^2\middle|\hat{S}=\hat{s}\right]\right] = E_{s|\theta}\left[\left(\frac{\partial}{\partial\theta}\log\sigma(s|\theta)\right)^2\right] = \mathcal{I}_F(\sigma(\theta))$$

Therefore, the Fisher information at almost every state decreases under garbling and so does the averaged Fisher information.

C.2 Fisher information cost is uniformly posterior separable

As mentioned in the model section, the cost can be defined over any action space A of finite size, and we start with

$$C_{FI}(p,\pi) = \int_\Theta \pi(t)\sum_{a\in A}\frac{[\frac{d}{dt}\sigma(a|t)]^2}{\sigma(a|t)}dt$$

where $\pi\in\Delta(\Theta)$ is a prior with full support, and $\sigma(a|t)$ denotes the probability of receiving a signal recommending action a at state t. We require $\sigma(a|t)>0$ for each $a\in A$ and almost all t, and the actions that are assigned with zero probability across all states are dropped from A.

We denote $\sigma(a)=\int_\Theta\sigma(a|t)\pi(t)dt$ as the ex-ante probability of choosing a, and $\sigma(\cdot|a)\in\Delta(\Theta)$ as the posterior over the states after receiving a recommendation for action a. More

specifically, we have $\sigma(t|a) = \frac{\sigma(a|t)\pi(t)}{\sigma(a)}$, and $\sum_a \sigma(a|t) = 1$. Also by chain rule

$$\frac{d}{dt}\sigma(t|a) = \frac{1}{\sigma(a)}[\frac{d}{dt}\sigma(a|t)\cdot\pi(t) + \frac{d}{dt}\pi(t)\cdot\sigma(a|t)]$$
$$\Rightarrow \frac{d}{dt}\sigma(a|t) = \frac{1}{\pi(t)}[\frac{d}{dt}\sigma(t|a)\cdot\sigma(a) - \frac{d}{dt}\pi(t)\cdot\sigma(a|t)]$$

We have:

$$\begin{aligned}
C_{FI}(p,\pi) &= \int_\Theta \pi(t)\sum_{a\in A}\frac{[\frac{d}{dt}\sigma(a|t)]^2}{\sigma(a|t)}dt \\
&= \int_\Theta \pi(t)\sum_{a\in A}\frac{1}{\sigma(a|t)}\frac{1}{\pi^2(t)}[\frac{d}{dt}\sigma(t|a)\cdot\sigma(a) - \frac{d}{dt}\pi(t)\cdot\sigma(a|t)]^2dt \\
&= \int_\Theta \sum_{a\in A}\frac{1}{\sigma(a|t)\pi(t)}[(\frac{d}{dt}\sigma(t|a))^2\sigma^2(a) - 2(\frac{d}{dt}\sigma(t|a))\sigma(a)\frac{d}{dt}\pi(t)\sigma(a|t) \\
&\quad + (\frac{d}{dt}\pi(t))^2\sigma^2(a|t)] \\
&= \int_\Theta \sum_a \sigma(a)\frac{(\frac{d}{dt}\sigma(t|a))^2}{\sigma(t|a)} - 2\frac{\frac{d}{dt}\pi(t)\cdot\frac{d}{dt}\sigma(t|a)}{\pi(t)}\sigma(t|a) + \frac{\sigma(a|t)(\frac{d}{dt}\pi(t))^2}{\pi(t)}dt \\
&= \int_\Theta \sum_a \sigma(a)\frac{(\frac{d}{dt}\sigma(t|a))^2}{\sigma(t|a)} - 2\frac{\sigma(a|t)(\frac{d}{dt}\pi(t))^2}{\pi(t)} - 2(\frac{d}{dt}\pi(t))(\frac{d}{dt}\sigma(a|t)) + \frac{\sigma(a|t)(\frac{d}{dt}\pi(t))^2}{\pi(t)}dt \\
&= \sum_a \sigma(a)\int_\Theta \frac{(\frac{d}{dt}\sigma(t|a))^2}{\sigma(t|a)} - \frac{(\frac{d}{dt}\pi(t)^2)}{\pi(t)}dt
\end{aligned}$$

The second-to-last equality uses the chain rule to expand the middle term

$$\frac{d}{dt}\sigma(t|a) = \frac{1}{\sigma(a)}[\frac{d}{dt}\sigma(a|t)\cdot\pi(t) + \frac{d}{dt}\pi(t)\cdot\sigma(a|t)]$$

The last equality holds because

$$\int_\Theta \sum_a (\frac{d}{dt}\pi(t))(\frac{d}{dt}\sigma(a|t))dt = \int_\Theta (\frac{d}{dt}\pi(t))(\frac{d}{dt}\sum_a \sigma(a|t))dt = \int_\Theta (\frac{d}{dt}\pi(t))(\frac{d}{dt}1)dt = 0$$

and

$$\int_\Theta \sum_a \frac{\sigma(a|t)(\frac{d}{dt}\pi(t))^2}{\pi(t)}dt = \int_\Theta \frac{(\frac{d}{dt}\pi(t))^2}{\pi(t)}dt = \sum_a \sigma(a)\int_\Theta \frac{(\frac{d}{dt}\pi(t))^2}{\pi(t)}dt$$

So, there exists a potential function $F : \Delta(\Theta)\mathbb{R}$ with the expression:

$$F(q) = \int_\Theta \frac{q'(t)^2}{q(t)}dt$$

Such a potential function doesn't depend on the prior and is convex because its integrand $\frac{(x')^2}{x}$ is convex. And we may express Fisher information cost as the expected difference between

the potential evaluated at the posterior and prior:

$$C_{FI}(p,\pi) = E_{\sigma|\pi}[F(\sigma(\cdot|a)) - F(\pi)] = \sum_a \sigma(a)F(\sigma(\cdot|a)) - F(\pi)$$

C.3 Justification for the normalizations

Our binary choice problem can be formally represented by the tuple $\langle \Theta, u, \pi, \lambda \rangle$ of state space, payoff, prior and scaling parameter:

$$\underset{p\in AC(\Theta,[0,1])}{\text{maximize}} \int_\Theta \pi(t)u(t)p(\theta) - \lambda\pi(t)\frac{(p'(\theta))^2}{p(\theta)(1-p(\theta))}dt.$$

Through appropriate rescaling and reparameterization of the payoff function $u(t)$, we can normalize the scaling parameter to $\lambda = 1$ normalize the state space to $[0,1]$. It is without loss of generality in the sense that the solution to any problem of the form $\langle \Theta, u, \pi, \lambda \rangle$ can be recovered from the solution to a normalized problem $\langle [0,1], \tilde{u}, \mathbf{1}, 1 \rangle$, for some appropriately transformed utility function $\tilde{u}$ and assume the prior π is uniform. We provide justification in the following.

Assume the scaling parameter is 1: Scaling the payoff function by a constant factor $\frac{1}{\lambda}$ results in an equivalent optimization problem, that is,

$$\begin{aligned}
&\underset{p\in AC(\Theta,[0,1])}{\text{maximize}} \int_\Theta \pi(t)u(t)p(\theta) - \lambda\pi(t)\frac{(p'(\theta))^2}{p(\theta)(1-p(\theta))}dt \\
\Leftrightarrow &\underset{p\in AC(\Theta,[0,1])}{\text{maximize}} \int_\Theta \pi(t)(\frac{1}{\lambda}u(t))p(\theta) - \pi(t)\frac{(p'(\theta))^2}{p(\theta)(1-p(\theta))}dt
\end{aligned}$$

the two problems yields the same solution.

Assume the state space is $[0,1]$: Given an original state space $[a,b]$, we can apply the transformation $y = \frac{t-a}{b-a}$. Solving for $p(\theta)$ in the original problem $\langle [a,b], u(t), \pi(t), \lambda \rangle$ is equivalent to first solving $\tilde{p}(y) := p((b-a)y+a)$ in the reparameterized problem $\langle [0,1], \tilde{u}(t), \tilde{\pi}(t), \frac{\lambda}{(b-a)^2} \rangle$, where $\tilde{u}(t) = u((b-a)y+a)$, $\tilde{\pi}(t) = \pi((b-a)y+a)$:

$$\begin{aligned}
&\underset{\tilde{p}\in AC([0,1],[0,1])}{\text{maximize}} \int_{[0,1]} [\tilde{\pi}(y)\tilde{u}(y)\tilde{p}(y) - \lambda\tilde{\pi}(y)\frac{\frac{1}{(b-a)^2}(\tilde{p}'(y))^2}{\tilde{p}(y)(1-\tilde{p}(y))}]d((b-a)y) \\
\Leftrightarrow &\underset{\tilde{p}\in AC([0,1],[0,1])}{\text{maximize}} \int_{[0,1]} \tilde{\pi}(y)\tilde{u}(y)\tilde{p}(y) - \frac{\lambda}{(b-a)^2}\tilde{\pi}(y)\frac{(\tilde{p}'(y))^2}{\tilde{p}(y)(1-\tilde{p}(y))}dy,
\end{aligned}$$

and then recovering $p(\theta)$ via $p(\theta) = \tilde{p}\left(\frac{t-a}{b-a}\right)$.

Assume the prior is uniform: Since π has full support, we can define $G(t) = \int_0^t \frac{1}{\pi(\tau)}\,d\tau$, which is strictly increasing and invertible. The reparameterization $y = G(t)$ transforms the original problem $\langle [a,b], u, \pi, \lambda \rangle$ into the problem

$\langle [0, G(b)], (\pi \circ G^{-1})^2 \cdot u \circ G^{-1}, \frac{\mathbf{1}_{[0,G(b)]}}{G(b)}, \lambda \rangle$, denote $\tilde{p}(y) := p(G^{-1}(y))$, we have:

$$(\tilde{p}(y))' = \frac{d}{dy} p(G^{-1}(y)) = \frac{d}{dt} p(\theta)|_{t=G^{-1}(y)} \frac{1}{\frac{d}{dt} G(t)|_{t=G^{-1}(y)}} = p'(\theta)\pi(t)|_{t=G^{-1}(y)}$$

$$\Rightarrow \underset{\tilde{p} \in AC([0,G(b)],[0,1])}{\text{maximize}} \int_0^{G(b)} [\pi(G^{-1}(y))u(G^{-1}(y))\tilde{p}(y) - \lambda \pi(G^{-1}(y)) \frac{(\frac{1}{\pi(G^{-1}(y))})^2 (\tilde{p}'(y))^2}{\tilde{p}(y)(1-\tilde{p}(y))}](\pi(G^{-1}(y))dy)$$

$$\Leftrightarrow \underset{\tilde{p} \in AC([0,G(b)],[0,1])}{\text{maximize}} \int_0^{G(b)} (\pi(G^{-1}(y)))^2 u(G^{-1}(y))\tilde{p}(y) - \lambda \frac{(\tilde{p}'(y))^2}{\tilde{p}(y)(1-\tilde{p}(y))} dy$$

Hence, we can first solve $\tilde{p}(y)$ from the problem $\langle [0, G(b)], (\pi \circ G^{-1})^2 \cdot u \circ G^{-1}, \frac{\mathbf{1}_{[0,G(b)]}}{G(b)}, \lambda \rangle$ and recover the original solution via $p(\theta) = \tilde{p}(G(t))$.